



ANGGA EKA SEPTIAN

DATA ANALYST PORTFOLIO

About

I bring an academic background in International Relations major in Economics and Statistics, combined with professional experience as a Junior Auditor at an accounting firm and as administrative and sales. I have strong English proficiency (TOEIC 830 / TOEFL 607) and hold a certification in data analysis tools, including Excel, SQL, Python, Tableau, and Power BI. My career spans both private and government sectors: Sales Marketing at PT Coca Cola Amatil Indonesia, Customer Service at PT Wahana Wisesa, Junior Auditor at KAP Supoyo Group, and Administrative Staff at the Government Narcotics Bureau of Pasuruan Regency. This cross-sector experience gives me a practical understanding of company reporting as well as government performance data.

I have also completed additional training to broaden these skills: a Data Analyst Ability Certificate from Refocus (2023) covering data processing, data visualization, and statistics; a training program for prospective auditors at Public Accounting Firm (KAP) Supoyo Group (2018-2019), covering accounting standards, audit procedures, sampling techniques, and risk identification; and Management training at PT Coca Cola Amatil (2017-2018), covering leadership, team management, and a comprehensive understanding of company operations from production and distribution to marketing and sales.

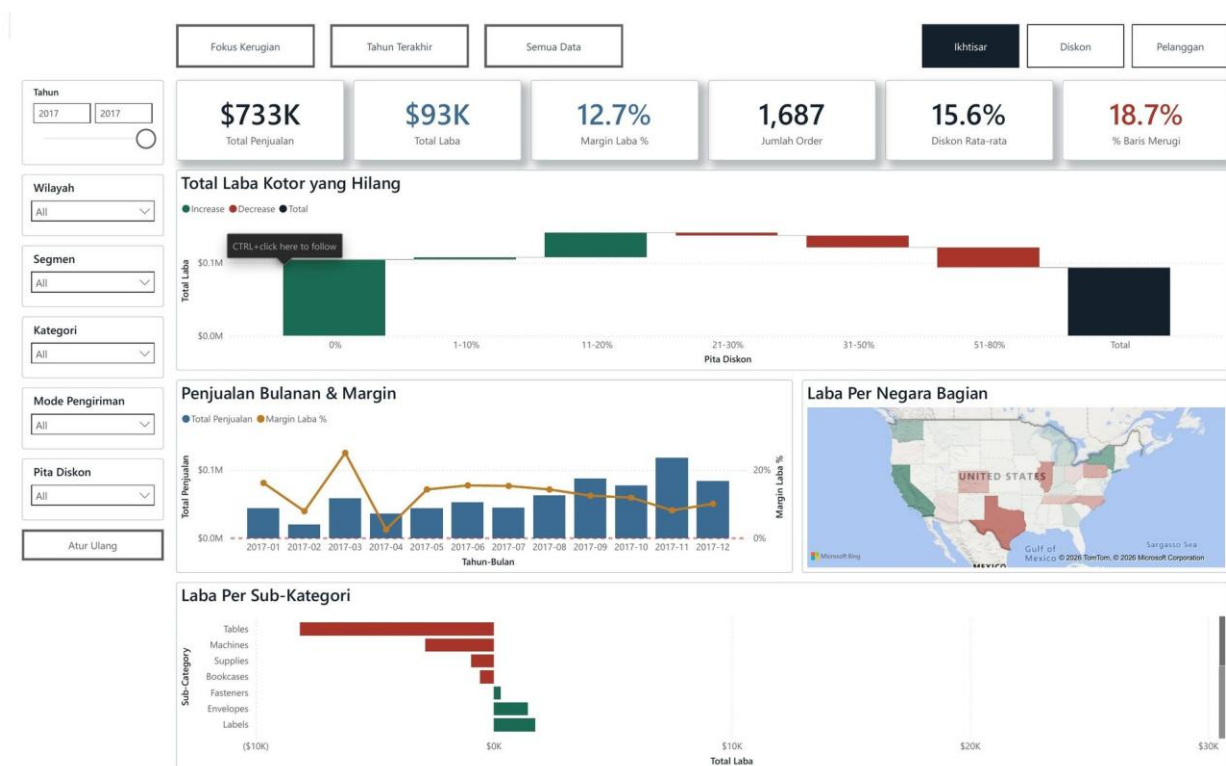
Project Examples

Project 1

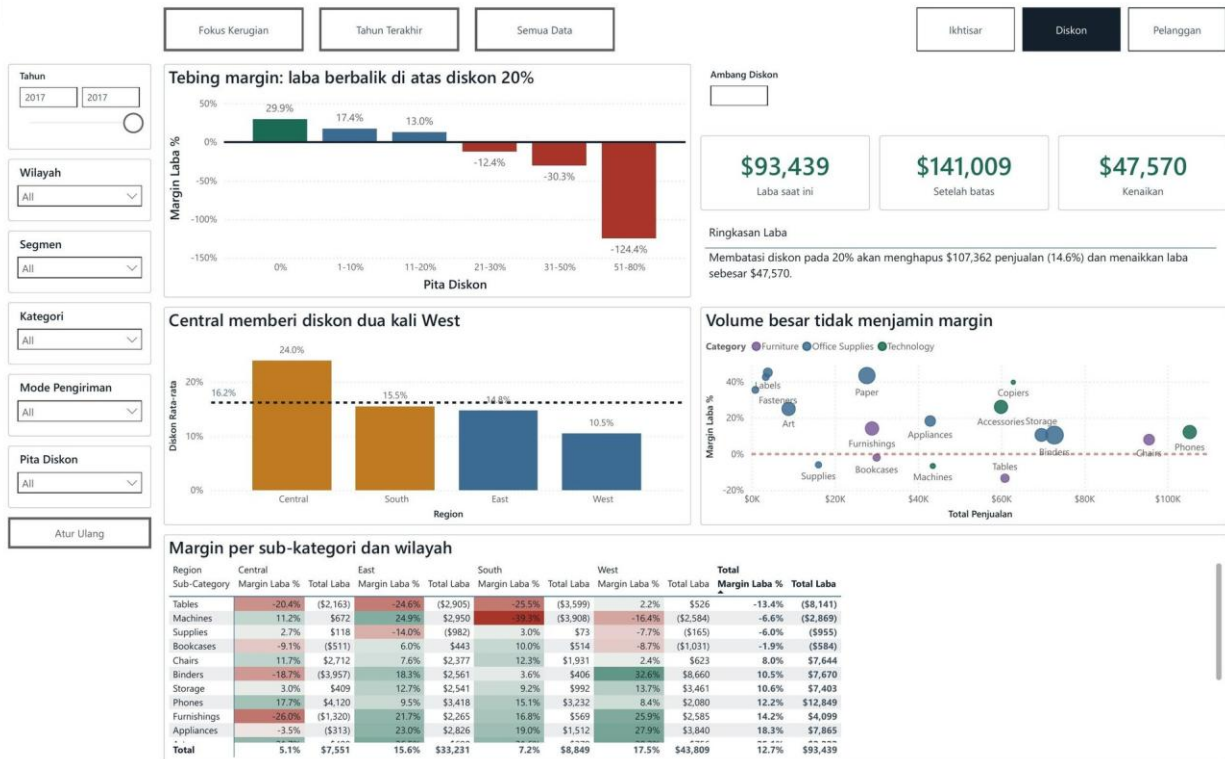
Discount & Profit-Leakage Analysis of Retail Sales Data (Superstore Dataset) — built in Power BI, Tableau, Excel, and Python (custom HTML/JS).

This project uses the widely-used Superstore transactional dataset (orders, sales, profit, and discount, 2014–2017) to diagnose where discounting is eroding profitability, segment customers and products by value, and quantify the profit impact of a discount policy change. The same analysis and business narrative were rebuilt on four platforms — Power BI, Tableau, Excel, and a hand-coded Python/HTML build — to demonstrate equivalent proficiency across DAX/Power Query, Tableau's calculated fields and dashboard actions, Excel's PivotTables, PivotCharts, and slicers, and a from-scratch pandas-to-JavaScript pipeline with no charting library at all.

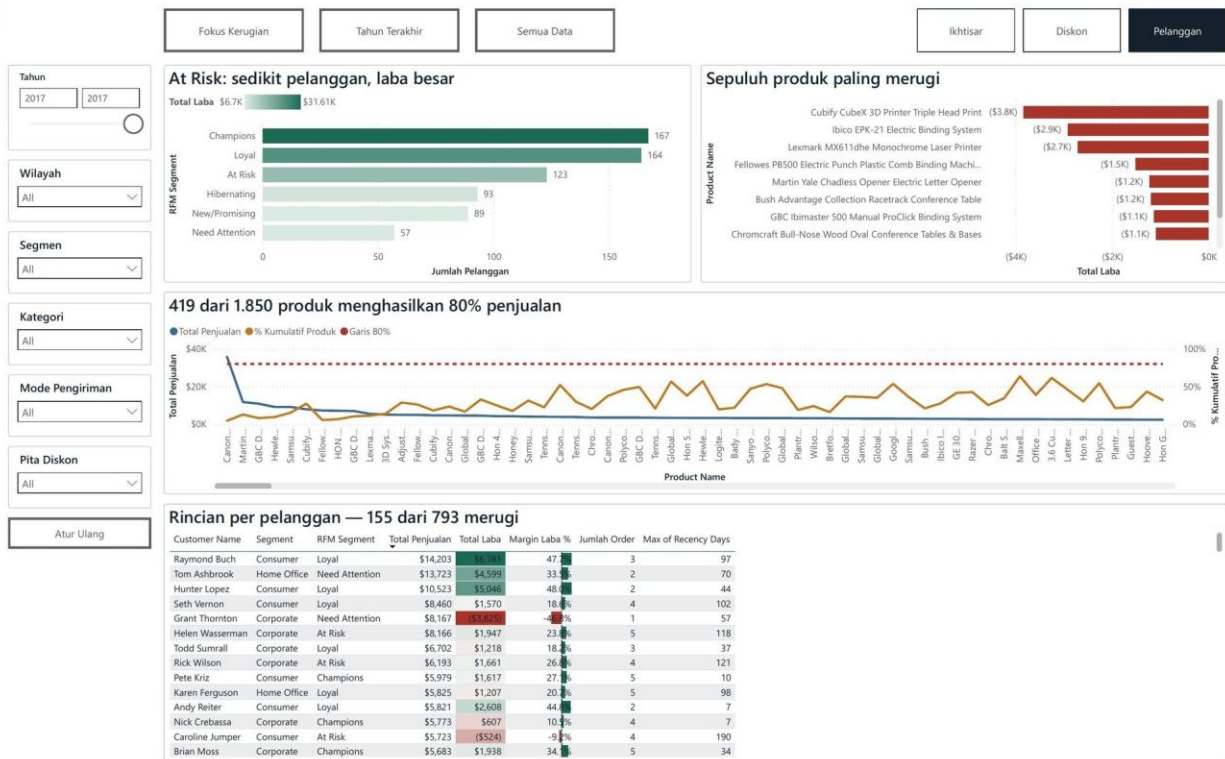
Power BI Dashboard



Overview (Ikhtisar): executive KPI cards (sales, profit, margin, orders, average discount, % loss-making transactions), a profit-leakage waterfall by discount band, the monthly sales & margin trend, and profit by state and sub-category — built with DAX measures and button-based tab navigation.

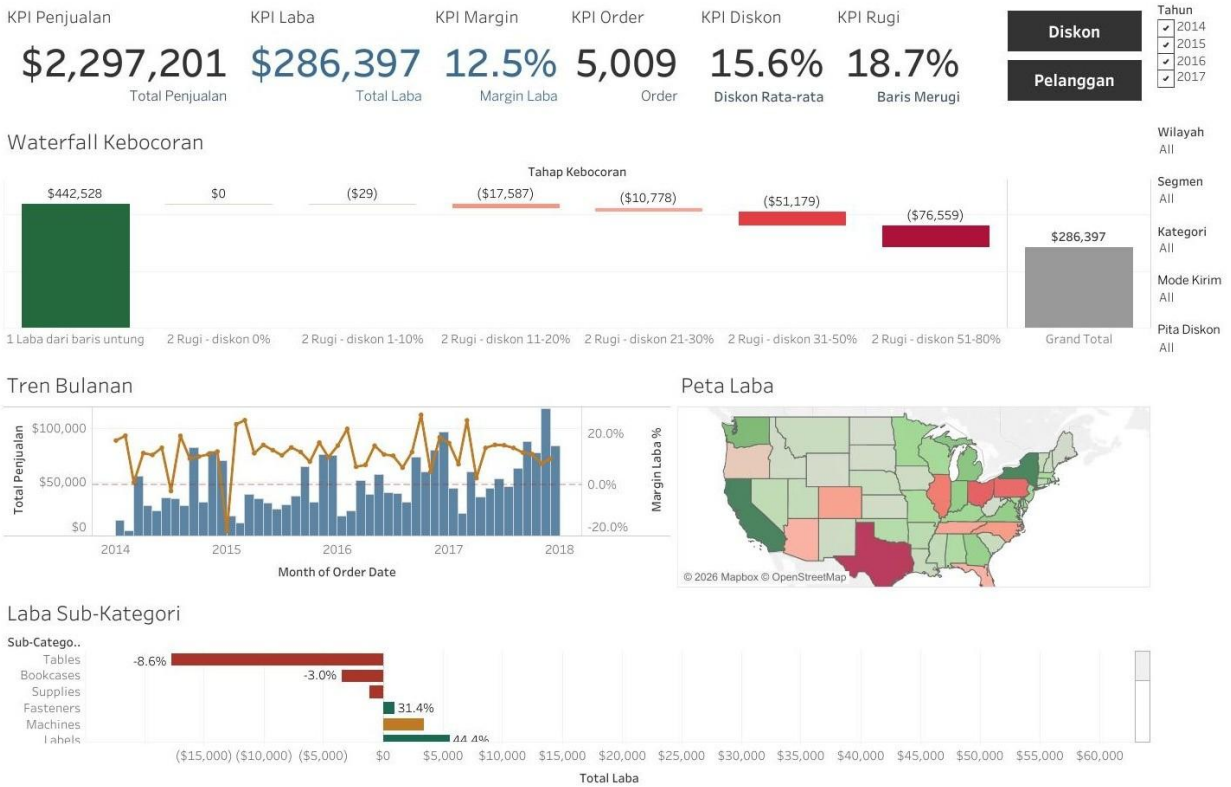


Discount (Diskon): the margin cliff where profit turns negative once discounts pass roughly 20%, a regional discount comparison (Central discounts about twice as much as West), a sales-volume-vs-margin scatter plot, and a margin matrix by sub-category and region — plus a what-if simulator showing that capping discounts at 20% would trade \$107K in sales for a \$47.6K gain in profit.



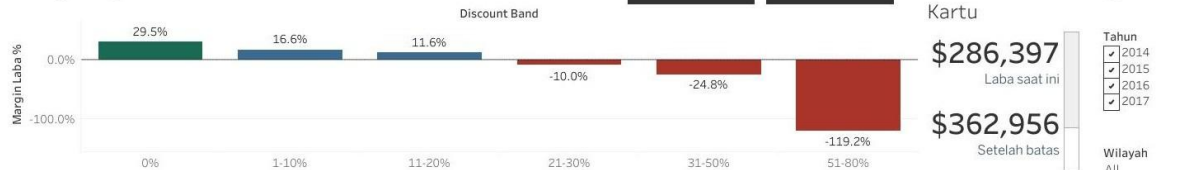
Customer (Pelanggan): RFM segmentation, the ten least profitable products, an 80/20 Pareto check (419 of 1,850 products drive 80% of sales), and a searchable customer-level profitability table flagging 155 of 793 customers as loss-making.

Tableau Dashboard



Overview (Ikhtisar), rebuilt in Tableau on the full 2014–2017 dataset: KPI cards, the same profit-leakage waterfall, monthly trend, and a profit-by-state map — using calculated fields and dashboard actions in place of DAX and bookmarks.

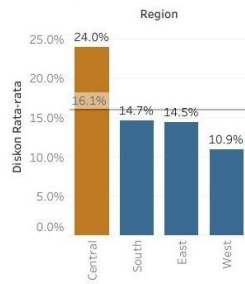
Tebing Margin



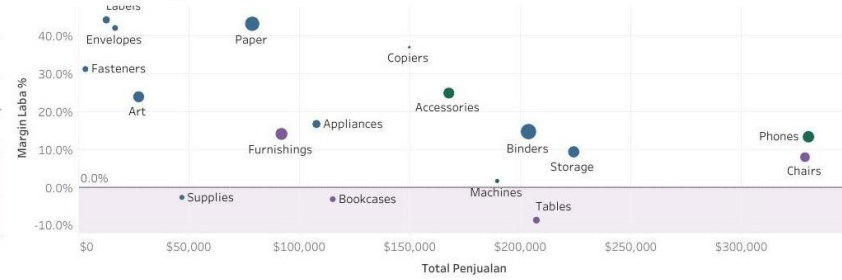
Simulator Narasi

Membatasi diskon pada 50% akan menghapus 64229 penjualan (2.8%) dan menaikkan laba sebesar 76559.

Diskon per Wilayah



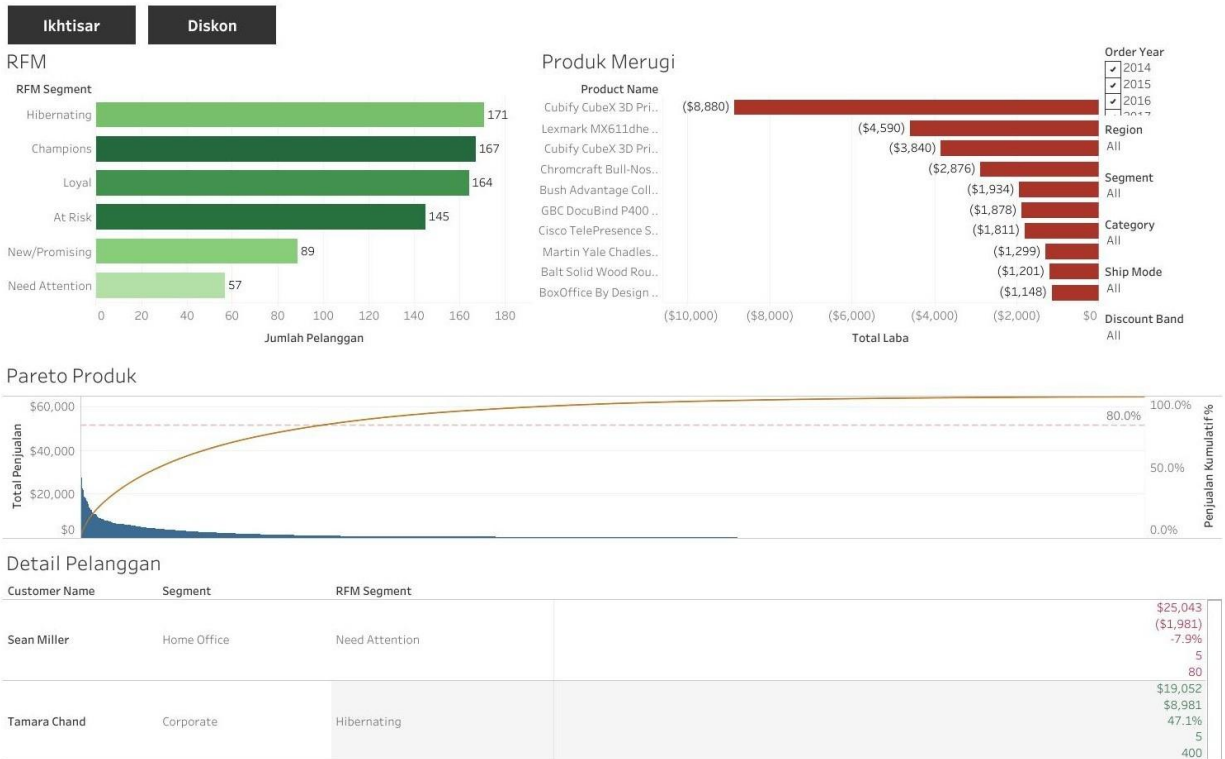
Sebaran Sub-Kategori



Matriks Margin

Sub-Category	Region			
	Central	East	South	West
Tables	-9.1%	-28.2%	-10.5%	1.7%
Bookcases	-8.3%	-2.7%	12.3%	-4.6%
Supplies	-7.0%	-10.7%	0.0%	3.5%
Fasteners	30.4%	32.2%	34.5%	29.8%
Machines	-5.5%	10.5%	-2.7%	-1.5%
Labels	43.8%	43.4%	44.2%	45.3%
Art	26.7%	26.4%	23.7%	26.0%

Discount (Diskon): the margin cliff at full-dataset scale (margin falls to -119% in the 51-80% discount band), regional discount comparison, sub-category margin scatter, a margin matrix, and a discount-threshold parameter modelling a 50% cap that recovers \$76.6K in profit.



Customer (Pelanggan): RFM segments, the ten most unprofitable products, and an 80/20 Pareto curve on product sales, all feeding into a drill-through detail view.

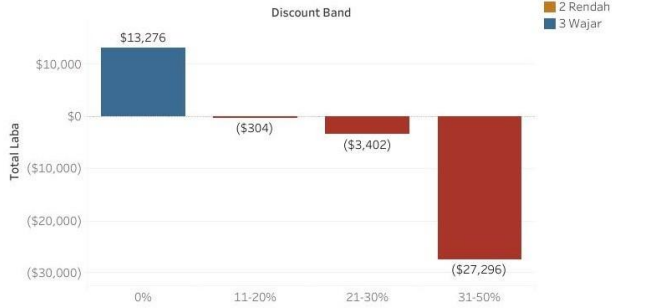
[← Kembali](#)

Detail Sub-Kategori

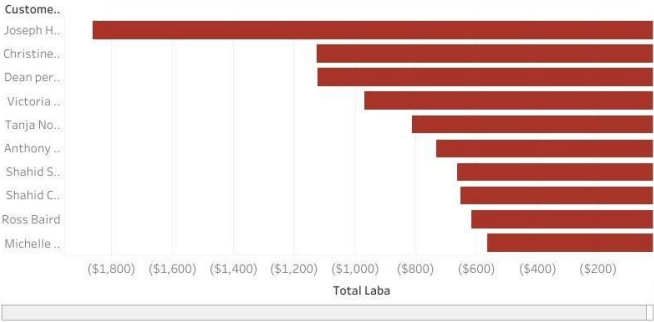
Detail Tren



Detail Diskon



Detail Pelanggan Rugi

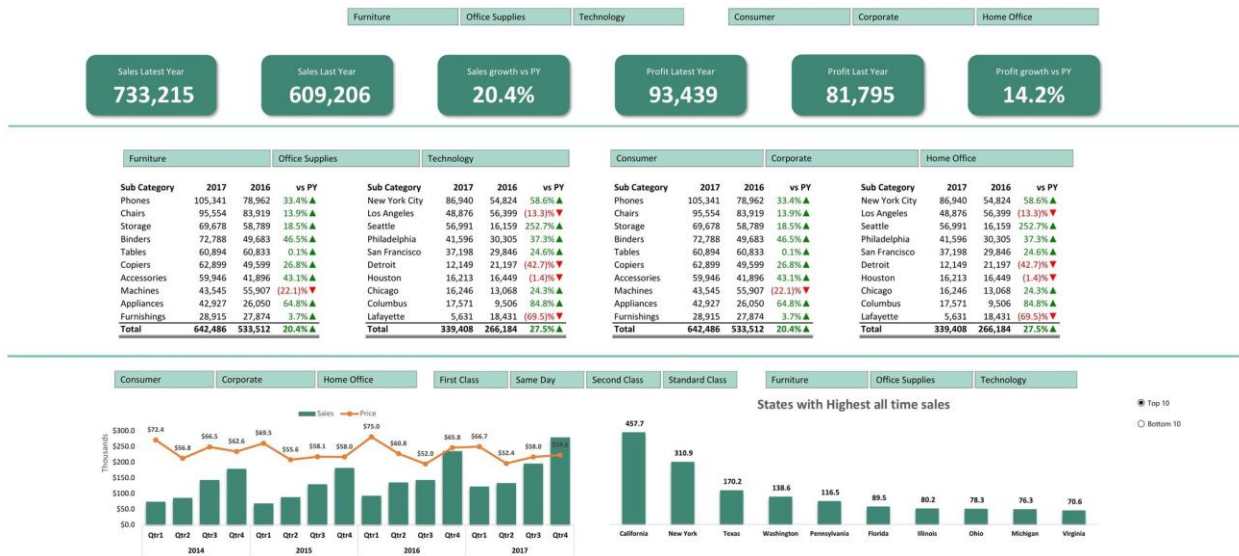


Detail Transaksi

Order ID	Year of O..	Customer..	State	Product Name	
CA-2014-100090	2014	Ed Braxton	California	Hon 2111 Invitation Series Corner Table	\$5023 20.0% (\$88)
CA-2014-100916	2014	Frank Hawley	Virginia	Hon 2111 Invitation Series Straight Table	\$5914 0.0% \$112
CA-2014-104283	2014	Liz MacKendrick	Mississippi	KI Adjustable-Height Table	\$861 0.0% \$22
CA-2014-105165	2014	Sam Zeldin	Texas	Riverside Furniture Oval Coffee Table, Oval End Table, End Table with Drawer	\$2011 30.0% (\$23)
CA-2014-105648	2014	Eva Jacobs	California	Bevis Oval Conference Table, Mahogany	\$6263 20.0%

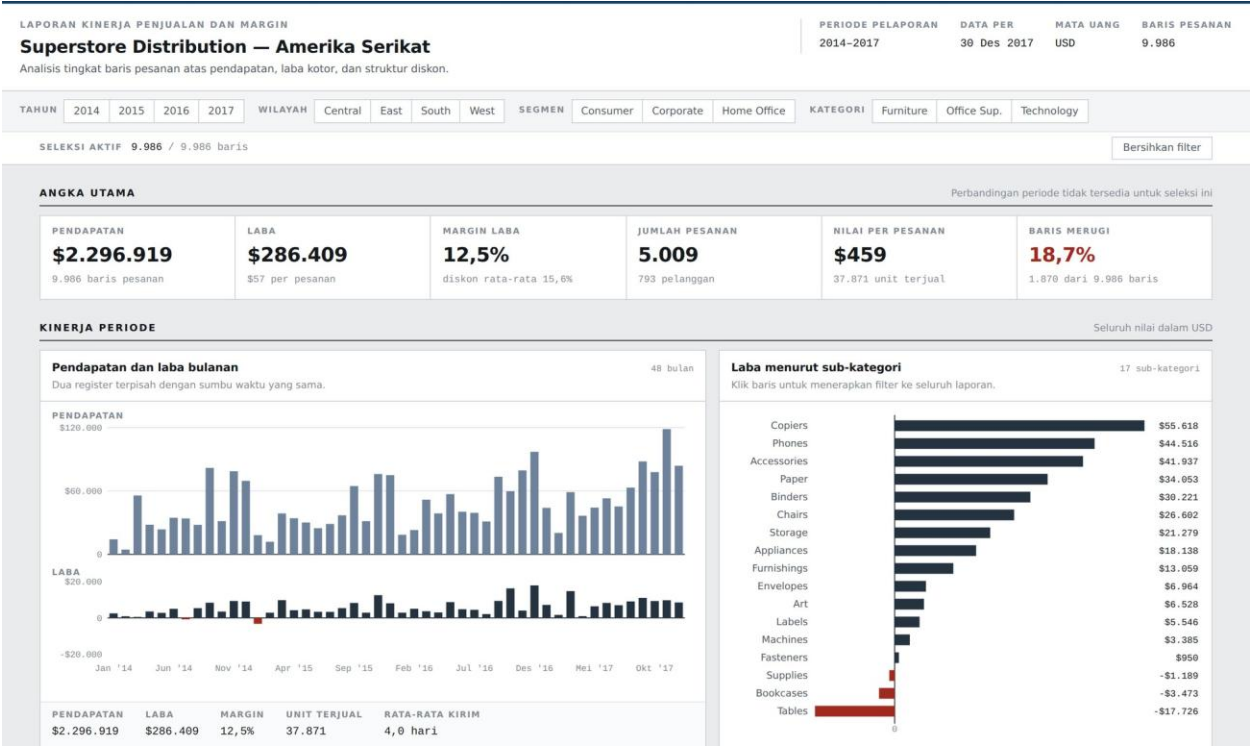
Sub-category drill-through: transaction-level detail reached from the Discount tab — monthly profit trend, profit by discount band, and the individual customers driving the largest losses.

Excel Dashboard

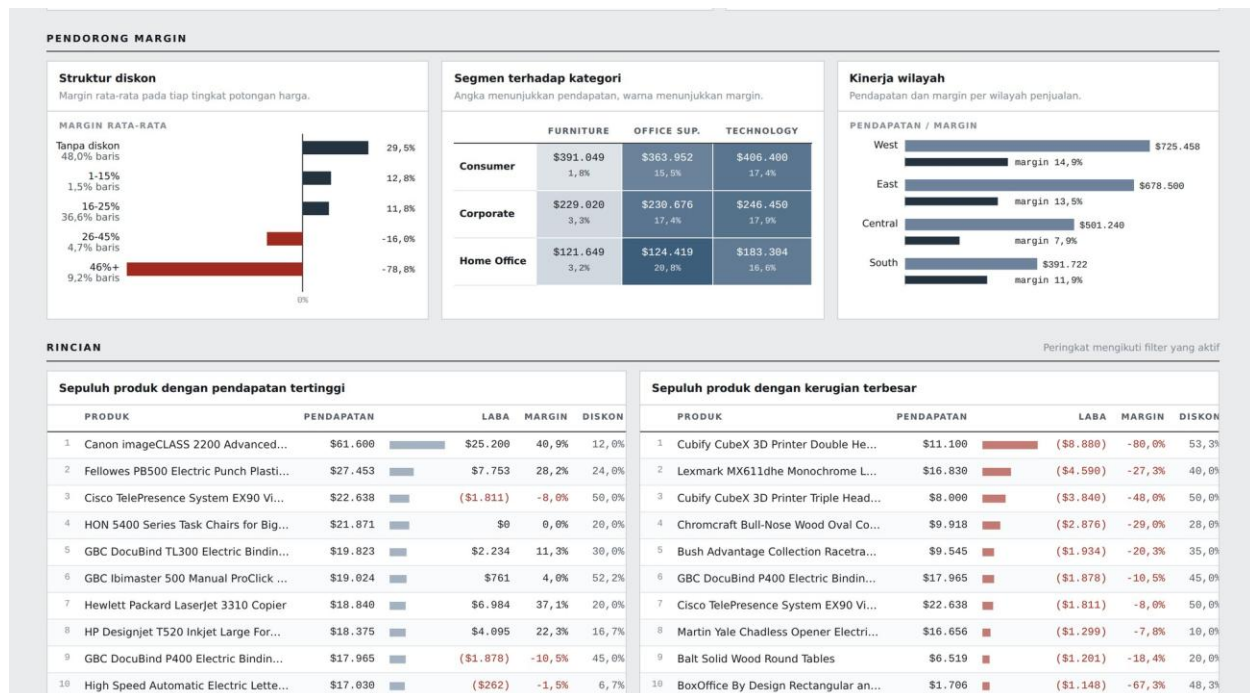


Excel (Ikhtisar): a slicer-driven dashboard built entirely with PivotTables, PivotCharts, and button-style filters for Category, Segment, and Ship Mode — on the same 2017 Superstore data used in the Power BI build. KPI cards track sales and profit against the prior year (sales +20.4%, profit +14.2%), Top-10 tables flag the fastest-moving sub-categories and cities (Appliances +64.8%, Seattle +252.7% vs. Lafayette -69.5%), a quarterly Sales-vs-Price trend runs 2014-2017, and a Top 10/Bottom 10 toggle ranks states by all-time sales, led by California.

Python + HTML/JS Dashboard



Angka Utama & Kinerja Periode: a single self-contained HTML file with zero JavaScript dependencies — no Chart.js, D3, or jQuery, every chart is hand-drawn SVG via the DOM API — sitting on top of a 13-step pandas cleaning pipeline (audited first: true duplicates removed, split line-items merged rather than dropped, dates and zero-padded ZIP codes fixed) over the full 2014-2017, 9,986-row dataset. KPI cards, a dual revenue/profit trend, and a click-to-filter sub-category ranking headline the report.



Pendorong Margin & Rincian: a discount-band margin analysis (profit swings from +29.5% with no discount to -78.8% at 46%+ discount), a segment-by-category matrix, region performance, and top-revenue/top-loss product tables — plus a sortable 49-state table and full methodology footnotes further down the report. All 9,986 rows ship inside the file itself for true cross-filtering and unique-order counts, compressed 5.8x with a custom dictionary + columnar JSON encoding, and every figure is independently re-verified against pandas and regression-tested with Playwright across 6 screen widths.

Key Insight & Recommendation

All four builds converge on the same finding at different scales: profit turns negative once a transaction's discount passes roughly 20%, a handful of sub-categories (Tables, Machines, Supplies, Bookcases) run at a loss overall, and a small share of customers and products accounts for a disproportionate share of the losses. That evidence supports capping discount approval at a threshold rather than cutting discounts across the board — the interactive simulator in each dashboard quantifies exactly how much profit that policy would recover. The Excel build adds a growth lens on the same data — sales and profit are both up year-over-year (+20.4% / +14.2%) — a reminder that any discount cap should be applied carefully so it doesn't slow the growth in the categories and cities driving it. The Python/HTML build reaches the same margin-cliff conclusion a fourth way, computed independently in pandas rather than any BI engine — reinforcing that the pattern is a property of the data, not an artifact of one tool's calculation.

Tools & Skills

Power BI (DAX measures, bookmark navigation, what-if parameters, drill-through), Tableau (calculated fields, parameters, dashboard actions, drill-through), Excel (PivotTables, PivotCharts, slicers), and Python (pandas for a 13-step data-auditing and cleaning pipeline, hand-written HTML/CSS/JS and SVG for a zero-dependency interactive report, Playwright for automated visual-regression testing), applied to discount/margin analysis, RFM customer segmentation, and Pareto analysis.

Python + HTML/JS Build — Code Walkthrough

The Python build was written from scratch rather than assembled in a BI tool, so the code is part of the deliverable. This walkthrough traces the pipeline end to end — audit, cleaning, payload encoding, dashboard, assembly, and automated testing — across roughly 1,275 lines in eight files. Each excerpt is labelled with its source file and line numbers. Source comments, originally written in Indonesian, have been translated and the multi-line audit-log calls collapsed to a single line; both changes were applied programmatically and verified by comparing abstract syntax trees before and after, so the logic shown is provably the logic that runs.

Pipeline inventory: `audit.py` (59 lines) and `audit2.py` (41) profile data quality; `clean_superstore.py` (188) runs thirteen cleaning steps behind five assertions; `build_payload.py` (69) encodes the cleaned data for the browser; `tpl_head.html` (349) and `tpl_js.html` (535) hold the document structure and all filtering, aggregation, and chart rendering; `assemble.py` (11) injects the payload into the template; `final.py` (35) drives the Playwright regression suite.

Stage 1 — Data Quality Audit

Nothing was cleaned before the defects were understood. The audit scripts only read and report; they never write. Two of the checks below matter more than they look: stray whitespace is invisible in a spreadsheet yet splits one product into two in any ranking, and a product identifier pointing at several different products silently invalidates it as a join key.

audit.py · lines 10–14 – duplicate detection at two levels

```
print("\n--- DUPLICATES ---")
print("Full-row duplicates:", df.duplicated().sum())
print("Duplicates ignoring Row ID:", df.drop(columns=['Row ID']).duplicated().sum())
key = ['Order ID', 'Product ID']
print("Duplicates on (Order ID + Product ID):", df.duplicated(subset=key).sum())
```

audit.py · lines 26–29 and 53–59 – whitespace and key-consistency checks

```
for c in df.select_dtypes(include='object').columns:
    s = df[c].astype(str)
    lead = (s != s.str.strip()).sum()
    if lead: print(f" {c}: {lead} values with stray whitespace")
```

```

print("\n--- KEY CONSISTENCY ---")
pp = df.groupby('Product ID')['Product Name'].nunique()
print("Product IDs with >1 name:", (pp>1).sum())
cc = df.groupby('Customer ID')['Customer Name'].nunique()
print("Customer IDs with >1 name:", (cc>1).sum())
sr = df.groupby('State')['Region'].nunique()
print("States in >1 region:", (sr>1).sum())

```

Stage 2 — Data Cleaning

Every step writes an entry to an audit trail, so the script produces a record of what changed rather than only a clean file.

clean_superstore.py · lines 5-19 – self-documenting log helper

```

import pandas as pd
import numpy as np
import json
import re

SRC = "/mnt/user-data/uploads/Sample_-_Superstore.csv"
LOG = []

def log(step, detail, before=None, after=None, impact="neutral"):
    LOG.append({
        "step": step, "detail": detail,
        "before": before, "after": after, "impact": impact
    })
    print(f"[{step}] {detail}" + (f" ({before} -> {after})" if before is not None else ""))

```

The source file is read as latin-1 rather than UTF-8, because latin-1 maps all 256 byte values and can never fail, so no row is dropped silently. Column names are then stripped of spaces and hyphens: “Sub-Category” breaks in several tools because the hyphen parses as a minus sign.

clean_superstore.py · lines 22-41 – encoding and column naming

```

# ----- 1. LOAD SOURCE FILE
raw = pd.read_csv(SRC, encoding="latin-1")
n0 = len(raw)
log("Load source file", ...) # audit-trail call, message elided

df = raw.copy()

# ----- 2. COLUMN NAMES
rename = {
    "Row ID": "line_id", "Order ID": "order_id", "Order Date": "order_date",
    "Ship Date": "ship_date", "Ship Mode": "ship_mode", "Customer ID": "customer_id",
    "Customer Name": "customer_name", "Segment": "segment", "Country": "country",
    "City": "city", "State": "state", "Postal Code": "postal_code", "Region": "region",
    "Product ID": "product_id", "Category": "category", "Sub-Category": "sub_category",
    "Product Name": "product_name", "Sales": "sales", "Quantity": "quantity",
    "Discount": "discount", "Profit": "profit",
}
df = df.rename(columns=rename)
log("Standardise column names", ...) # audit-trail call, message elided

```

Trimming alone is not enough for text: doubled spaces inside a product name split the same product into two entries. For dates the format string is explicit — left to inference, 03/04/2016 could be read as 3 April or 4 March, an error that raises no exception but corrupts every seasonal figure.

clean_superstore.py · lines 42-60 – text normalisation and date parsing

```
# ----- 3. TEXT NORMALISATION
text_cols = ["order_id", "ship_mode", "customer_id", "customer_name", "segment",
             "country", "city", "state", "region", "product_id", "category",
             "sub_category", "product_name"]
ws_before = sum((df[c].astype(str) != df[c].astype(str).strip()).sum() for c in text_cols)
for c in text_cols:
    df[c] = df[c].astype(str).strip()
    df[c] = df[c].str.replace(r"\s+", " ", regex=True) # collapse repeated spaces
log("Normalise text", ...) # audit-trail call, message elided

# ----- 4. DATES
df["order_date"] = pd.to_datetime(df["order_date"], format="%m/%d/%Y", errors="coerce")
df["ship_date"] = pd.to_datetime(df["ship_date"], format="%m/%d/%Y", errors="coerce")
bad_dates = df["order_date"].isna().sum() + df["ship_date"].isna().sum()
log("Parse dates", ...) # audit-trail call, message elided

neg_ship = (df["ship_date"] < df["order_date"]).sum()
log("Validate date logic", ...) # audit-trail call, message elided
```

pandas reads postal codes as integers, so 05408 becomes 5408; casting to a zero-padded string repairs 449 rows. For duplicates, line_id is excluded from the comparison because a row number is always unique and would mask every genuine duplicate.

clean_superstore.py · lines 61-72 – postal codes and exact duplicates

```
# ----- 5. POSTAL CODES
pz_before = (df["postal_code"].astype(str).str.len() < 5).sum()
df["postal_code"] = df["postal_code"].astype(int).astype(str).str.zfill(5)
log("Repair postal codes", ...) # audit-trail call, message elided

# ----- 6. EXACT DUPLICATES
dup_mask = df.drop(columns=["line_id"]).duplicated(keep="first")
dup_rows = int(dup_mask.sum())
dup_value = float(df.loc[dup_mask, "sales"].sum())
df = df[~dup_mask].copy()
log("Remove exact duplicates", ...) # audit-trail call, message elided
```

Fourteen further rows were the same product recorded twice within one order. These are merged, not deleted — deleting them would erase legitimate revenue. Discount is part of the grouping key rather than an aggregated column, because two lines carrying different discounts are genuinely different transactions. Summing sales rather than taking one of them leaves the revenue total unchanged.

clean_superstore.py · lines 73-89 – merging split order lines

```
# ----- 7. MERGE SPLIT ORDER LINES
key = ["order_id", "product_id", "discount"]
split_mask = df.duplicated(subset=key, keep=False)
n_split = int(split_mask.sum())
if n_split:
    static = ["order_date", "ship_date", "ship_mode", "customer_id", "customer_name",
             "segment", "country", "city", "state", "postal_code", "region",
             "category", "sub_category", "product_name"]
    agg = {c: "first" for c in static}
    agg.update({"line_id": "first", "sales": "sum", "quantity": "sum", "profit": "sum"})
    merged = df.groupby(key, as_index=False).agg(agg)
    df = merged[["line_id"] + key[:2] + ["order_date", "ship_date", "ship_mode",
                                       "customer_id", "customer_name", "segment", "country", "city", "state",
                                       "postal_code", "region", "category", "sub_category", "product_name",
                                       "sales", "quantity", "discount", "profit"]].copy()
log("Merge split order lines", ...) # audit-trail call, message elided
```

Thirty-two product identifiers point at genuinely different products. They are not repaired: the defect is in the source, and guessing the correct mapping would be inventing data. The rows are flagged and a surrogate key is derived instead.

clean_superstore.py · lines 90–103 – conflicting product identifiers

```
# ----- 8. CONFLICTING PRODUCT IDS
conflict = df.groupby("product_id")["product_name"].nunique()
n_conflict = int((conflict > 1).sum())
conflict_ids = set(conflict[conflict > 1].index)
df["product_id_conflict"] = df["product_id"].isin(conflict_ids)

def slug(s):
    return re.sub(r"^[a-z0-9]+", "-", s.lower()).strip("-")[:60]

df["product_key"] = df["product_id"] + "_" + df["product_name"].map(slug)
log("Handle conflicting product IDs", ...) # audit-trail call, message elided
```

Twelve derived columns are precomputed so the dashboard never recalculates them. `order_month` is stored as YYYY-MM text because that string sorts lexicographically in exactly chronological order, letting JavaScript sort months without parsing dates. The first discount band starts at -0.01 rather than 0 so a discount of exactly zero cannot fall into the wrong bucket.

clean_superstore.py · lines 104–120 – derived columns and discount bands

```
# ----- 9. DERIVED COLUMNS
df["order_year"] = df["order_date"].dt.year
df["order_quarter"] = df["order_date"].dt.year.astype(str) + "-Q" + df["order_date"].dt.quarter.astype(str)
df["order_month"] = df["order_date"].dt.strftime("%Y-%m")
df["order_month_name"] = df["order_date"].dt.strftime("%b")
df["ship_delay_days"] = (df["ship_date"] - df["order_date"]).dt.days
df["profit_margin"] = (df["profit"] / df["sales"]).round(4)
df["is_loss"] = df["profit"] < 0
df["unit_price"] = (df["sales"] / df["quantity"]).round(2)
df["discount_pct"] = (df["discount"] * 100).round(0).astype(int)
bands = [(-0.01, 0.001, "Tanpa diskon"), (0.001, 0.15, "1-15%"), (0.15, 0.25, "16-25%"),
         (0.25, 0.45, "26-45%"), (0.45, 1.01, "46%+")]
df["discount_band"] = pd.cut(df["discount"], bins=[b[0] for b in bands] + [1.01],
                             labels=[b[2] for b in bands], right=True, include_lowest=True)
df["discount_band"] = df["discount_band"].astype(str)
log("Add derived columns", ...) # audit-trail call, message elided
```

1,870 rows are loss-making and 349 lose more than the full sale value. All of them track discounts of 50–80 percent, so they are business fact rather than input error and they are kept — they are the central finding of the analysis. Five assertions then guard the output: if any fails the script stops and writes nothing, because producing no output beats producing quietly wrong output.

clean_superstore.py · lines 125–140 – outlier review and validation gate

```
# ----- 11. OUTLIERS (INSPECTED, NOT REMOVED)
loss_rows = int(df["is_loss"].sum())
extreme = int((df["profit_margin"] < -1).sum())
log("Inspect outliers", ...) # audit-trail call, message elided

# ----- 12. FINAL VALIDATION
assert df["sales"].min() > 0, "sales must be positive"
assert df["quantity"].min() > 0, "quantity must be positive"
assert df["discount"].between(0, 1).all(), "discount must fall in 0-1"
assert df["order_date"].notna().all(), "order_date must not be null"
assert df.isna().sum().sum() == 0, "no null values permitted"
log("Final validation", ...) # audit-trail call, message elided

df = df.sort_values("order_date").reset_index(drop=True)
df["line_id"] = np.arange(1, len(df) + 1)
```

Stage 3 — Payload Encoding

Cross-filtering and distinct order counts cannot be derived from pre-aggregated tables — unique orders do not sum across subtotals — so all 9,986 rows travel inside the HTML file. As ordinary JSON that is 3,767 KB. Three techniques together bring it to 653 KB: dictionary encoding, a columnar layout, and delimited strings in place of JSON arrays.

build_payload.py · lines 1-31 – dictionary encoding

```
"""Build the compact dashboard payload (columnar layout + dictionary encoding)."""
import pandas as pd, numpy as np, json

df = pd.read_pickle("/home/claude/clean.pkl")

def dict_encode(series, order=None):
    vals = order if order is not None else sorted(series.unique().tolist())
    idx = {v: i for i, v in enumerate(vals)}
    return vals, series.map(idx).astype(int).tolist()

months = sorted(df["order_month"].unique().tolist())
regions, r_code = dict_encode(df["region"])
states, s_code = dict_encode(df["state"])
segments, seg_code = dict_encode(df["segment"])
cats, c_code = dict_encode(df["category"])
subs, sub_code = dict_encode(df["sub_category"])
ships, sh_code = dict_encode(df["ship_mode"], ["Same Day", "First Class", "Second Class", "Standard Class"])
bands, b_code = dict_encode(df["discount_band"], ["Tanpa diskon", "1-15%", "16-25%", "26-45%", "46%+"])
prods, p_code = dict_encode(df["product_name"])
custs, cu_code = dict_encode(df["customer_name"])
orders, o_code = dict_encode(df["order_id"])

m_code = df["order_month"].map({m: i for i, m in enumerate(months)}).astype(int).tolist()

def j(arr):
    return ",".join(str(x) for x in arr)
```

Where the order of a dimension carries meaning it is passed explicitly, so the values are not sorted alphabetically. Each column is then joined into a single comma-delimited string, removing the per-element quotes and brackets that JSON would otherwise repeat 9,986 times.

build_payload.py • lines 32-59 – payload assembly

```
payload = {
    "meta": {
        "rows": int(len(df)),
        "dateMin": df["order_date"].min().strftime("%Y-%m-%d"),
        "dateMax": df["order_date"].max().strftime("%Y-%m-%d"),
        "orders": int(df["order_id"].nunique()),
        "customers": int(df["customer_id"].nunique()),
        "products": int(df["product_name"].nunique()),
        "salesTotal": float(df["sales"].round(2).sum()),
        "profitTotal": float(df["profit"].round(2).sum()),
    },
    "dim": {
        "month": months, "region": regions, "state": states, "segment": segments,
        "category": cats, "sub": subs, "ship": ships, "band": bands,
        "product": prods, "customer": custs, "order": orders,
    },
    "f": {
        "m": j(m_code), "r": j(r_code), "s": j(s_code), "g": j(seg_code),
        "c": j(c_code), "sc": j(sub_code), "sh": j(sh_code), "b": j(b_code),
        "p": j(p_code), "cu": j(cu_code), "o": j(o_code),
        "sal": j(df["sales"].round(2).tolist()),
        "pro": j(df["profit"].round(2).tolist()),
        "qty": j(df["quantity"].tolist()),
        "dsc": j(df["discount_pct"].tolist()),
        "dly": j(df["ship_delay_days"].tolist()),
    },
}
```

Stage 4 — Dashboard

The interface uses no JavaScript libraries — no charting framework, no component library. Every chart is SVG built through the DOM API, which guarantees the file works offline and keeps full control over presentation. Colour, type, and spacing are defined once as CSS custom properties, and because the JavaScript refers to the same tokens, changing one value restyles the charts, the tables, and the variance indicators together.

tpl_head.html • lines 11-21 – design tokens

```
:root{
  --ink:#15181c; --ink2:#545b64; --ink3:#848b94;
  --paper:#e9ebed; --surface:#ffffff; --surface2:#f4f5f7; --surface3:#fafbfc;
  --line:#ccd1d6; --line2:#e2e5e8;
  --bar:#6e829b; --barL:#b4c0cb;
  --pos:#25323f; --neg:#a02a20; --negL:#e9d4d1;
  --sel:#123a5e;
  --sans:'IBM Plex Sans',ui-sans-serif,system-ui,-apple-system,'Segoe UI',Arial,sans-serif;
  --cond:'IBM Plex Sans Condensed','IBM Plex Sans',ui-sans-serif,system-ui,sans-serif;
  --mono:'IBM Plex Mono',ui-monospace,'SF Mono',Menlo,Consolas,monospace;
}
```

On load, each delimited column is parsed once into a typed array. Typed arrays are laid out contiguously in memory with constant-time access, which matters because all 9,986 rows are rescanned every time a filter changes.

tpl_js.html · lines 7-24 – parsing into typed arrays

```
/* ===== column parsing ===== */
const iarr = s => { const p = s.split(','); a = new Int32Array(p.length);
  for (let i = 0; i < p.length; i++) a[i] = +p[i]; return a; };
const farr = s => { const p = s.split(','); a = new Float64Array(p.length);
  for (let i = 0; i < p.length; i++) a[i] = +p[i]; return a; };

const F = {
  m: iarr(D.f.m), r: iarr(D.f.r), s: iarr(D.f.s), g: iarr(D.f.g), c: iarr(D.f.c),
  sc: iarr(D.f.sc), sh: iarr(D.f.sh), b: iarr(D.f.b), p: iarr(D.f.p),
  cu: iarr(D.f.cu), o: iarr(D.f.o), sal: farr(D.f.sal), pro: farr(D.f.pro),
  qty: iarr(D.f.qty), dsc: iarr(D.f.dsc), dly: iarr(D.f.dly)
};

const N = F.m.length;
const YEARS = [...new Set(DIM.month.map(m => m.slice(0, 4)))].sort();
const monthYear = DIM.month.map(m => YEARS.indexOf(m.slice(0, 4)));
F.y = new Int32Array(N);
for (let i = 0; i < N; i++) F.y[i] = monthYear[F.m[i]];
```

Two number formats are used deliberately. Charts take the minus sign, since position relative to the zero line already communicates direction; tables take accounting parentheses, which read faster down a dense column of figures. Locale formatting produces 2.296.919 and 12,5% for an Indonesian audience.

tpl_js.html · lines 25-36 – number formatting

```
/* ===== number formatting ===== */
const nf = new Intl.NumberFormat('id-ID');
const nf1 = new Intl.NumberFormat('id-ID', { minimumFractionDigits: 1, maximumFractionDigits: 1 });
const int = n => nf.format(Math.round(n));
const money = n => (n < 0 ? '-' : '') + '$' + nf.format(Math.round(Math.abs(n)));
const acct = n => { const r = Math.round(n); return r < 0 ? '($' + nf.format(-r) + ')' : '$' + nf.format(r); };
const pct1 = n => nf1.format(n * 100) + '%';
const pp = n => nf1.format(Math.abs(n) * 100) + ' pp';
const MON = ['Jan', 'Feb', 'Mar', 'Apr', 'Mei', 'Jun', 'Jul', 'Agu', 'Sep', 'Okt', 'Nov', 'Des'];
const mLabel = k => { const [y, mm] = DIM.month[k].split('-'); return MON[+mm - 1] + " " + y.slice(2); };
const dLabel = iso => { const [y, m, d] = iso.split('-'); return +d + ' ' + MON[+m - 1] + ' ' + y; };
```

Filter state is held in Sets so membership tests are constant-time, and an empty Set means “no filter applied” rather than “nothing matches”, hence the size check before each lookup. The year-over-year comparison is the subtle part: it must ignore the year filter, since that is what is being shifted, while still honouring every other filter — which is why `passNonYear` is factored out and shared. If any month in the selection has no counterpart twelve months earlier, the function returns null and the dashboard shows context instead of a misleading figure.

tpl_js.html · lines 37-66 – filter state and the year-over-year baseline

```
/* ===== filter state ===== */
const ST = { year: new Set(), region: new Set(), segment: new Set(), category: new Set(), sub: null };

function passNonYear(i) {
  if (ST.region.size && !ST.region.has(F.r[i])) return false;
  if (ST.segment.size && !ST.segment.has(F.g[i])) return false;
  if (ST.category.size && !ST.category.has(F.c[i])) return false;
  if (ST.sub !== null && F.sc[i] !== ST.sub) return false;
  return true;
}

function activeRows() {
  const out = new Int32Array(N); let n = 0;
  const fy = ST.year.size;
  for (let i = 0; i < N; i++) {
    if (fy && !ST.year.has(F.y[i])) continue;
    if (!passNonYear(i)) continue;
    out[n++] = i;
  }
  return out.subarray(0, n);
}

/* comparison rows: same month range, shifted back twelve months */
function priorRows(rows) {
  const want = new Set(); let ok = true;
  for (let k = 0; k < rows.length; k++) { const q = F.m[rows[k]] - 12; if (q < 0) { ok = false; break; } want.add(q); }
  if (!ok || !want.size) return null;
  const out = new Int32Array(N); let n = 0;
  for (let i = 0; i < N; i++) { if (want.has(F.m[i]) && passNonYear(i)) out[n++] = i; }
  return n ? out.subarray(0, n) : null;
}
```

createElementNS is required: an element made with the ordinary createElement is an HTML element and will not render as SVG. Charts are drawn at real pixel width rather than scaled from a fixed viewBox, so text never stretches. The diverge helper solves a problem the first version got wrong — allocating a fixed share of width to the negative side made large losses overflow their labels. Sizing both sides in proportion to their peaks on one shared scale also makes the picture honest: on the discount chart the negative side takes 73% of the width because the losses really are that much larger than the gains.

tpl_js.html · lines 96-122 – SVG helpers and diverging-scale allocation

```
const NS = 'http://www.w3.org/2000/svg';
function S(tag, at, parent) {
  const e = document.createElementNS(NS, tag);
  for (const k in at) if (at[k] !== null && at[k] !== undefined) e.setAttribute(k, at[k]);
  if (parent) parent.appendChild(e);
  return e;
}

function initSvg(el, h) {
  el.innerHTML = '';
  const w = Math.max(el.clientWidth || el.parentNode.clientWidth - 26, 250);
  el.setAttribute('viewBox', `0 0 ${w} ${h}`);
  el.setAttribute('width', w); el.setAttribute('height', h);
  return { w, h };
}

const NICE = [1, 1.2, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10];
const niceMax = v => { if (v <= 0) return 1;
  const e = Math.pow(10, Math.floor(Math.log10(v))), f = v / e;
  return (NICE.find(s => f <= s + 1e-9) || 10) * e; };
const CW = 6.3;
const fitW = (strs, min, pad) => Math.max(min, Math.max(...strs.map(s => s.length)) * CW + pad);
function diverge(vals, x0, aw) {
  const mp = Math.max(0, ...vals), mn = Math.max(0, ...vals.map(v => -v));
  const tot = (mp + mn) || 1, k = aw / tot;
  return { zx: x0 + mn * k, k };
}

const POS = 'var(--pos)', NEGC = 'var(--neg)', BAR = 'var(--bar)';
```

Variance is expressed in two units. Currency and counts move in relative percent; margin and the loss-line share move in percentage points, because a margin going from 12.0% to 12.7% is +0.7 points, not +5.8%. The higherBetter flag inverts the favourable and adverse colouring for metrics where an increase is bad, such as the proportion of loss-making lines.

tpl_js.html · lines 141-148 – variance calculation

```
function deltaCell(cur, prev, kind, higherBetter) {
  if (prev === null) return null;
  let ch, txt;
  if (kind === 'pp') { ch = cur - prev; txt = pp(ch); }
  else { if (!prev) return null; ch = (cur - prev) / Math.abs(prev); txt = pct1(Math.abs(ch)); }
  const up = ch >= 0, fav = higherBetter ? up : !up;
  return `<span class="${fav ? 'fav' : 'adv'}">${up ? '▲' : '▼'} ${txt}</span><span>thn sb1m.</span>`;
}
```

One chart renderer in full, as a representative of the six. It computes its own row height, derives the value-column width from the longest number currently displayed, positions the zero line by proportional allocation, and wires click, keyboard, and hover behaviour onto every bar.

tpl_js.html · lines 226-261 – profit-by-sub-category chart, complete

```
function renderSub(rows) {
  const el = document.getElementById('cSub');
  const gr = groupBy(rows, F.sc, DIM.sub.length);
  const list = DIM.sub.map((n, i) => ({ n, i, p: gr.pro[i], s: gr.sal[i], c: gr.cnt[i], l: gr.loss[i] }))
    .filter(d => d.c > 0).sort((a, b) => b.p - a.p);
  document.getElementById('subTag').textContent = list.length + ' sub-kategori';
  const RH = 20, T = 6, B = 18, LW = 94;
  const H = Math.max(list.length * RH + T + B, 56), g = initSvg(el, H);
  if (!list.length) return;
  const VW = Math.min(fitW(list.map(d => money(d.p)), 50, 12), g.w * 0.34);
  const aw = Math.max(g.w - LW - VW, 40);
  const { zx, k } = diverge(list.map(d => d.p), LW, aw);

  S('line', { x1: zx, y1: T - 2, x2: zx, y2: H - B + 4, class: 'zero' }, el);
  list.forEach((d, i) => {
    const y = T + i * RH, w = d.p * k, neg = d.p < 0;
    const active = ST.sub === d.i, dimmed = ST.sub !== null && !active;
    const grp = S('g', { class: 'bar-int' + (dimmed ? ' dim' : ''), tabindex: 0, role: 'button',
      'aria-label': `${d.n}, laba ${money(d.p)}`, }, el);
    S('rect', { x: 0, y, width: g.w, height: RH, fill: active ? 'var(--surface2)' : 'transparent' }, grp);
    S('text', { x: LW - 8, y: y + RH / 2 + 3.5, 'text-anchor': 'end',
      class: active ? 'lbl-b' : 'lbl' }, grp).textContent = d.n;
    S('rect', { x: neg ? zx + w : zx, y: y + 4, width: Math.max(Math.abs(w), 1.2), height: RH - 8,
      fill: neg ? NEGC : POS }, grp);
    S('text', { x: g.w - 2, y: y + RH / 2 + 3.5, 'text-anchor': 'end', class: 'val',
      fill: neg ? NEGC : 'var(--ink)' }, grp).textContent = money(d.p);
    bindTip(grp, `<div class="tt">${d.n}</div>` + throw('Laba', money(d.p)) +
      throw('Pendapatan', money(d.s)) + throw('Margin', d.s ? pct1(d.p / d.s) : '-') +
      throw('Baris merugi', `${int(d.l)} dari ${int(d.c)}`));
    const go = () => { ST.sub = ST.sub === d.i ? null : d.i; render(); };
    grp.addEventListener('click', go);
    grp.addEventListener('keydown', e => { if (e.key === 'Enter' || e.key === ' ') { e.preventDefault(); go(); } });
  });
  S('text', { x: zx, y: H - 5, 'text-anchor': 'middle' }, el).textContent = '0';
}
```

Stage 5 — Assembly

Eleven lines, one of which is load-bearing. The payload is injected into a script tag, so a product name containing the characters that close that tag would truncate the document and break the entire dashboard; escaping them as unicode escapes keeps JSON.parse returning the identical

string. The assertion makes a missing placeholder fail loudly instead of shipping an empty dashboard.

assemble.py · complete file

```
import json, os
payload = open('/home/claude/payload.json').read()
# escape so the payload cannot terminate the <script> tag
payload = payload.replace('<', '\\u003c').replace('>', '\\u003e').replace('&', '\\u0026')
html = open('/home/claude/tpl_head.html').read() + open('/home/claude/tpl_js.html').read()
assert '__PAYLOAD__' in html
html = html.replace('__PAYLOAD__', payload)
os.makedirs('/mnt/user-data/outputs', exist_ok=True)
out = '/mnt/user-data/outputs/superstore_dashboard.html'
open(out, 'w', encoding='utf-8').write(html)
print(f"{out} -> {os.path.getsize(out)/1024:.0f} KB")
```

Stage 6 — Automated Testing

Rendering tests prove the dashboard is not broken; they do not prove the numbers are right. Both were checked. Text overflow is measured from real SVG bounding boxes rather than judged from a screenshot, and clipping inside cards is detected by comparing scroll width against client width. The suite covers six viewport widths from 390 to 1512 pixels against five filter combinations — thirty layout permutations per run. Separately, every published figure was reconciled against independent pandas calculations over the same cleaned data.

final.py · lines 19-24 – layout assertions

```
ov = pg.evaluate("()=>{const bad=[];document.querySelectorAll('svg.chart').forEach(s=>{
  const W=s.getAttribute('width');s.querySelectorAll('text').forEach(t=>{const bb=t.getBBox();
    if(bb.x<-1||bb.x+bb.width>W+1)bad.push(s.id+':'+t.textContent);});});return bad;}")
clip = pg.evaluate("()=>[...document.querySelectorAll('.kpi-v,.cell .c1,.pf-v')].filter(e=>e.scrollWidth>e.clientWidth+1).length")
hscroll = pg.evaluate("document.documentElement.scrollWidth>window.innerWidth+2")
print(f" {name:26s} overflow={str(ov or 'ok'):<22} clipped={clip} h-scroll={hscroll} err={errs or 'ok'}")
```

Five defects were caught this way, including sticky table footers floating over panel headings and a truncated final column. One suspected defect turned out to be an artefact of screenshot downscaling: a line that appeared to cross a panel title vanished at three times resolution, and `getComputedStyle` confirmed no element was there. Verifying a visual suspicion by measurement before changing code saved a pointless edit.

Project 2

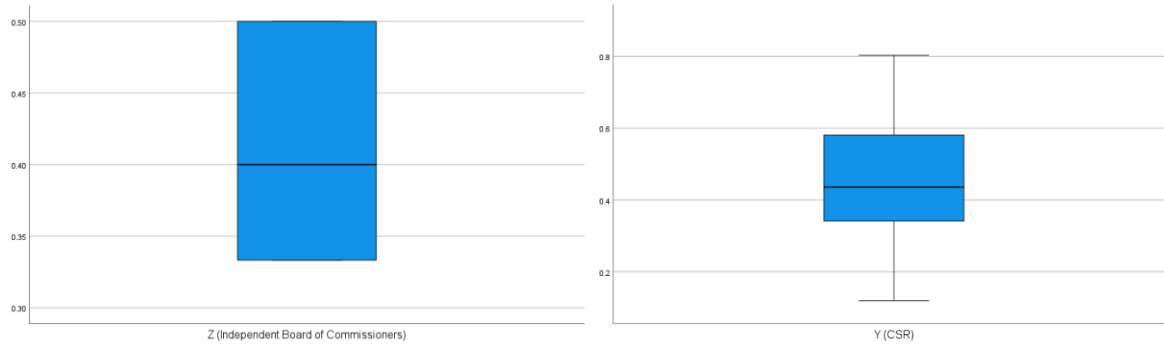
Analysis of the effect of Ownership Structure (foreign ownership and institutional ownership) on Corporate Social Responsibility (CSR), with the Independent Board of Commissioners as a moderating variable, in 30 consumer goods sector manufacturing companies listed on the IDX for the period 2021 to 2023.

periode	Company	X1 (Foreign Ownership)	X2 (Institutional Ownership)	Z (Independent Board of Commissioners)	Y (CSR)	X1*Z	X2*Z	Y_Transform
2021	AISA	0.1432903	0.57495323	0.33333333	0.478632	0.001214	-0.00947	.480238289801271
2022	AISA	0.0454067	0.63616589	0.4	0.692308	0.001327	-0.00213	.536207862847277
2023	AISA	0.0552941	0.73060565	0.5	0.803419	-0.00914	0.02441	.587223026425659
2021	ALTO	0	0.38651713	0.5	0.136752	-0.01402	-0.00597	.313313075143792
2022	ALTO	0	0.38651708	0.5	0.136752	-0.01402	-0.00597	.313313065723100
2023	ALTO	0	0.38651708	0.5	0.136752	-0.01402	-0.00597	.313313065723100
2021	BTEK	0	0.41591551	0.5	0.239316	-0.01402	-0.00338	.349127118736050
2022	BTEK	0	0.41591551	0.5	0.42735	-0.01402	-0.00338	.403657033265964
2023	BTEK	0.1302586	0.27010969	0.5	0.435897	-0.00252	-0.01625	.398469071819787
2021	BUDI	0	0.57847752	0.33333333	0.444444	0.012444	-0.00974	.457589794934421
2022	BUDI	0	0.59759289	0.33333333	0.461538	0.012444	-0.01124	.463948086130086
2023	BUDI	0	0.59759289	0.33333333	0.461538	0.012444	-0.01124	.463948086130086
2021	CAMP	0	0.83874257	0.33333333	0.461538	0.012444	-0.03014	.481622712441984
2022	CAMP	0	0.83874257	0.5	0.538462	-0.01402	0.033958	.523188555152218
2023	CAMP	0	0.83874257	0.5	0.444444	-0.01402	0.033958	.495923597887261
2021	CEKA	0.049563	0.87020336	0.33333333	0.410256	0.00856	-0.03261	.473550708551513
2022	CEKA	0	0.87020336	0.33333333	0.487179	0.012444	-0.03261	.491364471578027
2023	CEKA	0	0.87020336	0.33333333	0.709402	0.012444	-0.03261	.555808916022472
2021	CLEO	0	0.81374167	0.33333333	0.288136	0.012444	-0.02818	.429503485263664
2022	CLEO	0	0.7688	0.33333333	0.350282	0.012444	-0.02466	.444232166536813
2023	CLEO	0	0.7688	0.33333333	0.378531	0.012444	-0.02466	.452424256932293
2021	COCO	0.6111529	0.05423688	0.5	0.316384	0.039942	-0.03531	.402082542071621

Data Cleaning (outlier, missing values, converting into data panel) using excel and python.

*** data obtained from financial reports of consumer goods sector manufacturing companies listed on the IDX for the period 2021 to 2023 (<https://www.idx.co.id/>).**





Box Plot Test to search for outlier in each variable for data cleaning purpose.

Descriptive Statistics

	N	Statistic	Minimum	Maximum	Mean		Std. Deviation
					Statistic	Std. Error	
X1 (Foreign Ownership)	90		0.000	0.893	0.159	0.026	0.244
X2 (Institutional Ownership)	90		0.000	0.870	0.454	0.029	0.270
Z (Independent Board of Commissioners)	90		0.333	0.500	0.412	0.008	0.077
Y (CSR)	90		0.120	0.803	0.452	0.017	0.162
Valid N (listwise)	90						

Descriptive Statistics using SPSS.

When evaluating data quality, particularly for regression analysis and statistical modeling, classical assumption tests serve as critical indices for data quality assessment. So, it is necessary to conduct classical assumption tests first, which include: data normality tests, multicollinearity tests, heteroscedasticity tests and autocorrelation tests. These tests help ensure that the statistical models will produce reliable, unbiased results (Hair et al., 2019: 376-378).

One-Sample Kolmogorov-Smirnov Test

			Unstandardized Residual
N			90
Normal Parameters ^{a,b}	Mean		0.0000000
	Std. Deviation		0.04760221
Most Extreme Differences	Absolute		0.059
	Positive		0.059
	Negative		-0.045
Test Statistic			0.059
Asymp. Sig. (2-tailed) ^c			0.200
Monte Carlo Sig. (2-tailed) ^e	Sig.		0.613
	99% Confidence Interval	Lower Bound	0.601
		Upper Bound	0.626

a. Test distribution is Normal.

b. Calculated from data.

c. Lilliefors Significance Correction.

d. This is a lower bound of the true significance.

e. Lilliefors' method based on 10000 Monte Carlo samples with starting seed

NORMALITY TEST

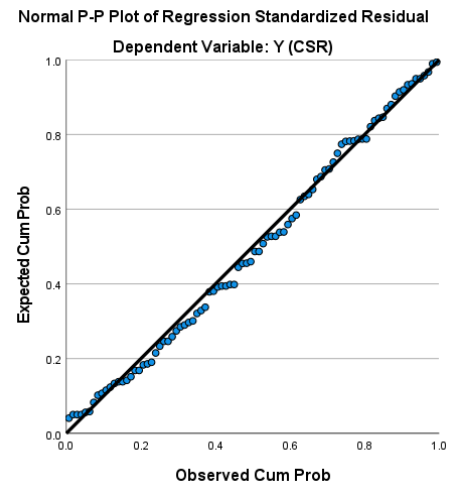
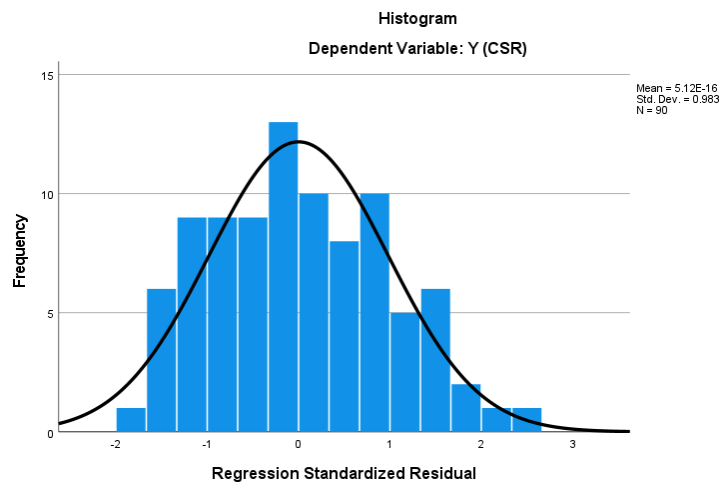
Data is said to be normally distributed if:

Asymp. Sig. (2-tailed) > 0.05

(NORMALLY Distributed)

Asymp. Sig. (2-tailed) < 0.05

(Not NORMALLY Distributed)



Normality Test using SPSS.

AUTOCORRELATION TEST

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.732a	0.536	0.520	0.04843	0.899

a. Predictors: (Constant), Z (Independent Board of Commissioners), X1 (Foreign Ownership), X2 (Institutional Ownership)

b. Dependent Variable: Y (CSR)

Durbin-Watson Test Conditions: $DW < DL$ or $D > 4 - DL$ (**AUTOCORRELATION occurs**)
 $DU < DW < 4 - DU$ (**No AUTOCORRELATION Occurs**)
 $DL < DW < DU$ or $4 - DU < DW < 4 - DL$ (**No Conclusion**)

Durbin-Watson Test:

DW Value	0.899
DU Value (K=3 and N= 90)	1.7261
4 - DU Value	2.2739

Conclusion

AUTOCORRELATION occurs

AUTOCORRELATED TEST WITH COCHRANE ORCUTT METHOD

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.692a	0.479	0.461	0.04038	1.891

a. Predictors: (Constant), Z (Independent Board of Commissioners), X1 (Foreign Ownership), X2 (Institutional Ownership)

b. Dependent Variable: Y (CSR)

Durbin-Watson Test:

DW Value	1.891
DU Value (K=3 and N= 90)	1.7261
4 - DU Value	2.2739

Conclusion

No AUTOCORRELATION occurs

Autocorrelation Test Using Two Different Methods in SPSS.

MULTICOLLINEARITY TEST

It is said that the data does not show symptoms of multicollinearity if:

Tolerance Value > 0.1 and VIF Value < 10 (**No Multicollinearity Symptoms Occur**)

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	0.384	0.037		10.307	0.000		
	X1 (Foreign Ownership)	0.204	0.028	0.711	7.259	0.000	0.563	1.778
	X2 (Institutional Ownership)	0.208	0.026	0.803	8.074	0.000	0.545	1.835
	Z (Independent Board of Commissioners)	-0.143	0.071	-0.159	-2.033	0.045	0.884	1.132

a. Dependent Variable: Y (CSR)

Multicollinearity Test using SPSS.

HETEROSCEDASTICITY TEST

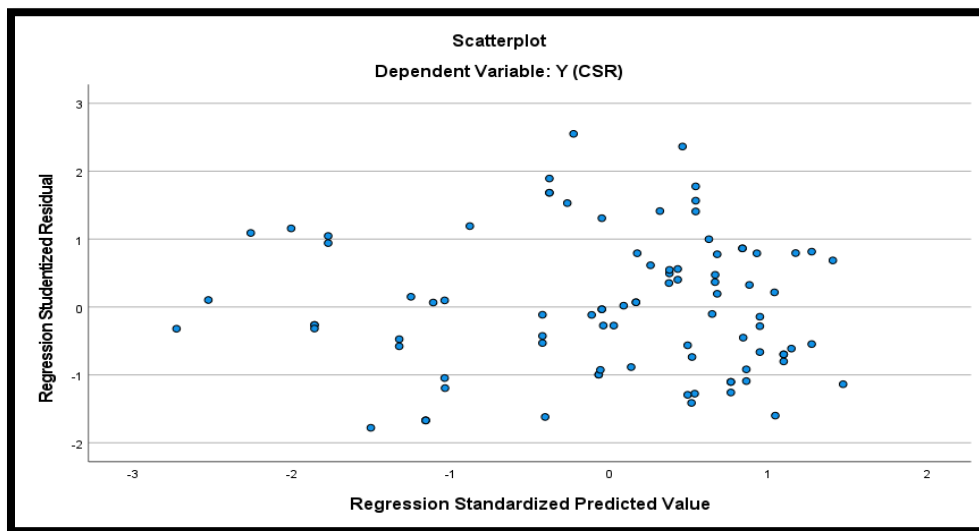
It is said that the data does not show symptoms of multicollinearity if:

Sig > 0.05 (**HETEROSKEDASTICITY Does Not Occur**)

Sig < 0.05 (**HETEROSKEDASTICITY occurs**)

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	0.047	0.021		2.246	0.027
	X1 (Foreign Ownership)	0.009	0.016	0.082	0.579	0.564
	X2 (Institutional Ownership)	-0.009	0.014	-0.087	-0.604	0.547
	Z (Independent Board of Commissioners)	-0.014	0.040	-0.039	-0.345	0.731

a. Dependent Variable: ABSRES



Heteroscedasticity Test Using SPSS

Hypothesis Testing and Data Analysis Conclusion

PARTIAL TEST (T TEST)

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficient	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	0.384	0.037		10.307	0.000
	X1 (Foreign Ownership)	0.204	0.028	0.711	7.259	0.000
	X2 (Institutional Ownership)	0.208	0.026	0.803	8.074	0.000
	Z (Independent Board of Commissioners)	-0.143	0.071	-0.159	-2.033	0.045

a. Dependent Variable: Y (CSR)

Notes :

It is said to have **a significant effect** if the Sig. value < 0.05

It is said to have **no effect** if the Sig. value > 0.05

CONCLUSION:

1. The Foreign Ownership variable (X1) has a t-test result > t table ($7.259 > 1.98761$) and is significant at 0.000, less than 0.05, so it can be concluded that Foreign Ownership (X1) has a significant effect on CSR (Y) with a positive effect direction.
2. The Institutional Ownership variable (X2) has a t-test result > t table ($8.074 > 1.98761$) and is significant at 0.000, less than 0.05, so it can be concluded that Institutional Ownership (X2) has a significant effect on CSR (Y) with a positive effect direction.
3. The Independent Board of Commissioners variable (Z) has a t-test result > t table ($-2.033 > -1.98761$) and is significant at 0.045, which is smaller than 0.05, so it is concluded that CSR (Y) has a significant effect on Institutional Ownership (X2) with a negative effect direction.

MODERATION TEST

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficient	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	0.408	0.036		11.238	0.000
	X1 (Foreign Ownership)	0.182	0.029	0.633	6.173	0.000
	X2 (Institutional Ownership)	0.191	0.025	0.741	7.583	0.000
	Z (Independent Board of Commissioners)	-0.161	0.067	-0.178	-2.407	0.018
	X1 (Foreign Ownership) * Z (Independent Board of Commissioners)	0.690	0.375	0.184	1.837	0.070
	X2 (Institutional Ownership) * Z (Independent Board of Commissioners)	1.126	0.323	0.328	3.487	0.001

a. Dependent Variable: Y (CSR)

CONCLUSION:

1. The influence of the relationship between the Independent Board of Commissioners (Z) as a Moderator on the relationship between Foreign Ownership (X1) and CSR (Y) obtained a calculated t value < t table ($1.837 < 1.98761$) and a significance of 0.070 greater than 0.05 so that it was concluded that the Independent Board of Commissioners cannot moderate the influence of the relationship between Foreign Ownership and CSR.

2. The influence of the relationship between the Independent Board of Commissioners (Z) as a Moderation on the relationship between Institutional Ownership (X2) and CSR (Y) obtained a calculated t value > t table ($3.487 > 1.98761$) and a significance of 0.001, less than 0.05, so it was concluded that the Independent Board of Commissioners can moderate the influence of the relationship between Institutional Ownership and CSR.

GOODNESS OF FIT MODEL

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.772a	0.597	0.573	0.046

a. Predictors: (Constant), X2 (Institutional Ownership) * Z (Independent Board of Commissioners), Z (Independent Board of Commissioners), X1 (Foreign Ownership) * Z (Independent Board of Commissioners), X2 (Institutional Ownership), X1 (Foreign Ownership)

b. Dependent Variable: Y (CSR)

CONCLUSION:

CSR (Y) is influenced by Foreign Ownership (X1) and Institutional Ownership (X2) with the Independent Board of Commissioners (Z) as Moderation, by 59.7%. While the other 40.3% is the influence that comes from other variables that are not research variables.

SIMULTANEOUS TEST (F TEST)

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	0.259	5	0.052	24.840	.000b
	Residual	0.175	84	0.002		
	Total	0.435	89			

a. Dependent Variable: Y (CSR)

b. Predictors: (Constant), X2 (Institutional Ownership) * Z (Independent Board of Commissioners), Z (Independent Board of Commissioners), X1 (Foreign Ownership) * Z (Independent Board of Commissioners), X2 (Institutional Ownership), X1 (Foreign Ownership)

Notes :

It is said to have **a significant effect** if the Sig. value < 0.05

It is said to have **no effect** if the Sig. value > 0.05

CONCLUSION:

It can be seen for the calculated F value = 24.840 to determine the calculated F value with a significance level of 5% with degrees of freedom, df = (n-k) or (90-2) and (k-1) or (2-1), df (90-2 = 88) and (2-1 = 1) the results obtained for the F table are 3.95. So the results of the calculated F > F table ($24.840 > 3.95$) with a sig value of ($0.000 < 0.05$) then simultaneously (together) the independent variables Foreign Ownership (X1), Institutional Ownership (X2) and Independent Board of Commissioners (Z) as Moderation have a significant effect on CSR (Y).

Project 3

Analysis of factors influencing the use of electronic payments among university students in Malaysia.

Gender	Age	Educational	Allowance	States	Study_Year	Knowledge	Usage	Application
Male	31 – 35 years old	Degree	RM1,500 and above	Kelantan	2	Yes	Yes	Others
Female	31 – 35 years old	Degree	RM1,500 and above	Kelantan	2	Yes	Yes	GrabPay
Female	Above 40 years old	Master	RM1,500 and above	Kelantan	2	Yes	Yes	MAE (maybank E-Wallet)
Female	25 – 30 years old	Master	RM1,500 and above	Wilayah Persekutuan	2	Yes	Yes	Touch 'n Go e-Wallet
Female	22 – 24 years old	Degree	RM501 - RM1,000	Selangor	3	Yes	Yes	GrabPay
Female	Below 18 years old	Diploma	RM500 or less	Kelantan	1	Yes	Yes	Touch 'n Go e-Wallet
Male	19 - 21 years old	Diploma	RM1,500 and above	Kelantan	5	Yes	Yes	Touch 'n Go e-Wallet
Female	19 - 21 years old	Degree	RM500 or less	Kedah	3	Yes	Yes	Touch 'n Go e-Wallet
Female	Below 18 years old	Diploma	RM501 - RM1,000	Selangor	1	Yes	Yes	MAE (maybank E-Wallet)
Male	19 - 21 years old	Degree	RM500 or less	Perak	3	Yes	Yes	Touch 'n Go e-Wallet
Male	19 - 21 years old	Degree	RM500 or less	Kelantan	2	Yes	Yes	GrabPay
Female	Below 18 years old	Diploma	RM500 or less	Kelantan	1	Yes	Yes	MAE (maybank E-Wallet)
Female	22 – 24 years old	Degree	RM501 - RM1,000	Selangor	3	Yes	Yes	Touch 'n Go e-Wallet
Female	22 – 24 years old	Degree	RM500 or less	Kelantan	3	Yes	Yes	Touch 'n Go e-Wallet
Female	22 – 24 years old	Degree	RM500 or less	Kelantan	3	Yes	Yes	Touch 'n Go e-Wallet
Female	22 – 24 years old	Degree	RM501 - RM1,000	Selangor	5	Yes	Yes	MAE (maybank E-Wallet)
Female	31 – 35 years old	Degree	RM501 - RM1,000	Kelantan	2	Yes	Yes	Touch 'n Go e-Wallet
Female	25 – 30 years old	Degree	RM500 or less	Kelantan	3	Yes	Yes	MAE (maybank E-Wallet)
Female	19 - 21 years old	Diploma	RM500 or less	Kelantan	3	Yes	Yes	Others
Female	19 - 21 years old	Diploma	RM500 or less	Selangor	2	Yes	Yes	Others
Female	19 - 21 years old	Degree	RM501 - RM1,000	Selangor	2	Yes	Yes	MAE (maybank E-Wallet)
Female	19 - 21 years old	Diploma	RM501 - RM1,000	Wilayah Persekutuan	2	Yes	Yes	Touch 'n Go e-Wallet
Female	22 – 24 years old	Degree	RM500 or less	Kelantan	3	Yes	Yes	Others
Female	36 – 40 years old	Degree	RM500 or less	Kelantan	2	Yes	Yes	Touch 'n Go e-Wallet
Female	19 - 21 years old	Diploma	RM500 or less	Kelantan	3	Yes	No	Others
Female	Below 18 years old	Diploma	RM500 or less	Kelantan	2	Yes	No	Touch 'n Go e-Wallet
Female	19 - 21 years old	Degree	RM501 - RM1,000	Selangor	4	Yes	Yes	Touch 'n Go e-Wallet

Sorting data based on set parameters on the questionnaire form. Data Cleaning (outlier, missing values, converting into data panel) using excel and python.

*** data used in this study was acquired through a quantitative examination. The main data collection tool in this study was a structured self-administered questionnaire survey that was adapted from past investigations and analysis (Manikandan & Jayakodi, 2017; Goh, 2017; Smolarczyk, 2018).**

The questionnaire has two sections: Section A and Section B. In Section A, it asked for demographic breakdown of the student's background. In Section B, students were asked to rate how much they agreed or disagreed with each statement using five Likert scales in order to provide information about their opinions on the influence of perceived usefulness, perceived ease of use, and subjective norm on E-Wallet acceptability. Scale "1" indicates a strong disagreement with the statement, while "5" indicates a strong agreement.

This study was conducted through an online survey using the google form questionnaire on the targeted focus group comprising of students. Then it is followed by collecting quantitative data and analysis with descriptive statistics using SPSS. The data were analysed using Statistical Structural Equation Model (SEM) using Partial Least Square (PLS) analysis to test the proposed hypotheses. The combined data analysis method was used to evaluate all of the data collected from the questionnaire.

Respondent Demographic		Frequency	Percent
Gender	Male	62	25
	Female	186	75
	Total	248	100
Age	Below 18 years old	32	12.9
	19 - 21 years old	123	49.6
	22 – 24 years old	74	29.8
	25 – 30 years old	9	3.6
	31 – 35 years old	6	2.4
	36 – 40 years old	2	0.8
	Above 40 years old	2	0.8
	Total	248	100
Education Qualification	Diploma	99	40
	Degree	139	56
	Master	10	4
	Total	248	100
Monthly Allowance	RM500 or less	198	79.8
	RM501 - RM1,000	31	12.5
	RM1,001 - RM1,500	2	0.8
	RM1,500 and above	17	6.9
	Total	248	100
States	Wilayah Persekutuan	10	4
	Kelantan	145	58.5
	Sabah	1	0.4
	Selangor	60	24.2
	Perak	24	9.7
	Terengganu	2	0.8
	Pahang	1	0.4
	Johor	1	0.4
	Pulau Pinang	3	1.2
	Kedah	1	0.4
	Total	248	100
Year of Study	1 Year	56	22.6
	2 Years	83	33.5
	3 Years	56	22.6
	4 Years	35	14.1
	5 Years and Above	18	7.3
	Total	248	100
Heard Digital Apps	Yes	248	100
	No	0	0
	Total	248	100

Used Digital Apps	Yes	221	89.1
	No	27	10.9
	Total	248	100
Digital Application	Touch 'n Go	155	62.5
	Boost	9	3.6
	GrabPay	19	7.7
	MAE (Maybank)	29	11.7
	Others	36	14.5
	Total	248	100

Gender					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Female	186	75.0	75.0	75.0
	Male	62	25.0	25.0	100.0
	Total	248	100.0	100.0	

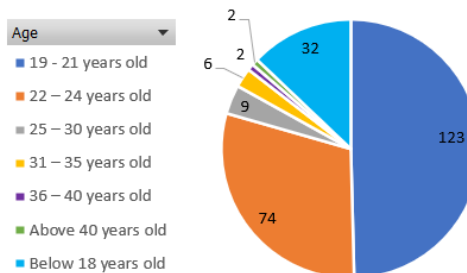
Age					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	19 - 21 years old	123	49.6	49.6	49.6
	22 - 24 years o	74	29.8	29.8	79.4
	25 - 30 years o	9	3.6	3.6	83.1
	31 - 35 years o	6	2.4	2.4	85.5
	36 - 40 years o	2	0.8	0.8	86.3
	Above 40 years ol	2	0.8	0.8	87.1
	Below 18 years ol	32	12.9	12.9	100.0
	Total	248	100.0	100.0	

Educational					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Degree	139	56.0	56.0	56.0
	Diplom	99	39.9	39.9	96.0
	Master	10	4.0	4.0	100.0
	Total	248	100.0	100.0	

Allowance					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	RM1,001 - RM1,500	2	0.8	0.8	0.8
	RM1,500 and above	17	6.9	6.9	7.7
	RM500 or less	198	79.8	79.8	87.5
	RM501 - RM1,000	31	12.5	12.5	100.0
	Total	248	100.0	100.0	

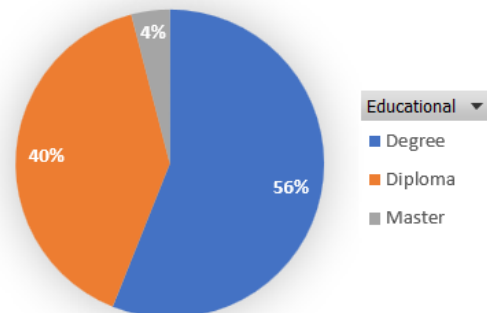
Age	Count of Age
19 - 21 years old	123
22 - 24 years old	74
25 - 30 years old	9
31 - 35 years old	6
36 - 40 years old	2
Above 40 years old	2
Below 18 years old	32
Grand Total	248

Count of Age



Education	Count of Educational
Degree	139
Diploma	99
Master	10
Grand Total	248

Count of Educational



States					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Johor	1	0.4	0.4	0.4
	Kedah	1	0.4	0.4	0.8
	Kelantan	145	58.5	58.5	59.3
	Pahang	1	0.4	0.4	59.7
	Perak	24	9.7	9.7	69.4
	Pulau Pi	3	1.2	1.2	70.6
	Sabah	1	0.4	0.4	71.0
	Selangor	60	24.2	24.2	95.2
	Terengga	2	0.8	0.8	96.0
	Wilayah	10	4.0	4.0	100.0
	Total	248	100.0	100.0	

Study_Year					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	56	22.6	22.6	22.6
	2	83	33.5	33.5	56.0
	3	56	22.6	22.6	78.6
	4	35	14.1	14.1	92.7
	5	18	7.3	7.3	100.0
	Total	248	100.0	100.0	

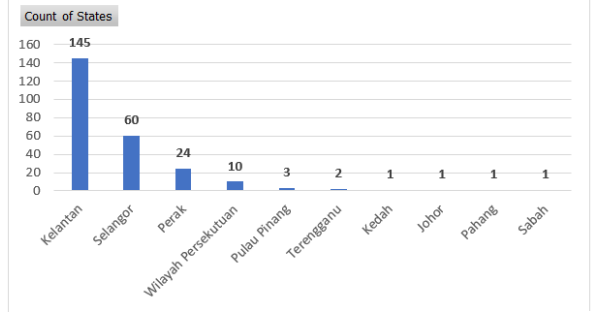
Knowledge					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Yes	248	100.0	100.0	100.0

Usage					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	27	10.9	10.9	10.9
	Yes	221	89.1	89.1	100.0
	Total	248	100.0	100.0	

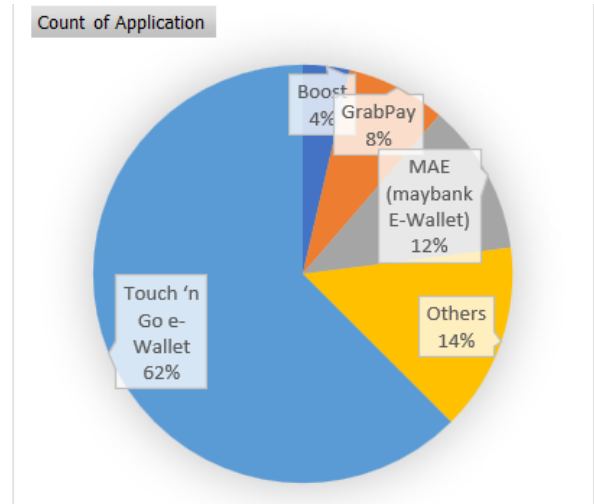
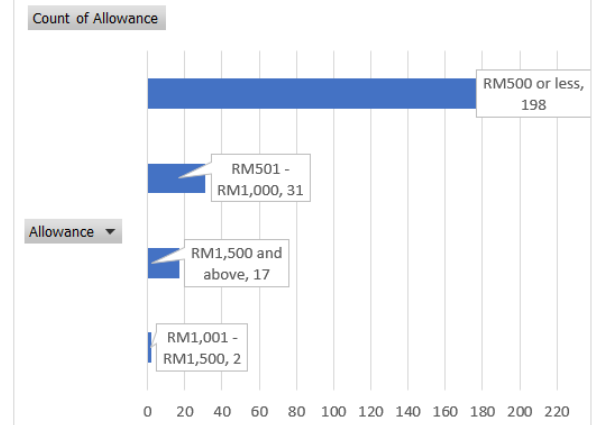
Application					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Boost	9	3.6	3.6	3.6
	GrabPa	19	7.7	7.7	11.3
	MAE (Maybank)	29	11.7	11.7	23.0
	Others	36	14.5	14.5	37.5
	Touch 'n Go	155	62.5	62.5	100.0
	Total	248	100.0	100.0	

Application	Count of Application
Boost	9
GrabPay	19
MAE (maybank E-Wallet)	29
Others	36
Touch 'n Go e-Wallet	155
Grand Total	248

States	Count of States
Kelantan	145
Selangor	60
Perak	24
Wilayah Persekutuan	10
Pulau Pinang	3
Terengganu	2
Kedah	1
Johor	1
Pahang	1
Sabah	1
Grand Total	248



Allowance	Count of Allowance
RM1,001 - RM1,500	2
RM1,500 and above	17
RM501 - RM1,000	31
RM500 or less	198
Grand Total	248



Descriptive statistics results using SPSS.

PU1	PU2	PU3	PU4	PE1	PE2	PE3	PE4	PE5	SN1	SN2	SN3	DA1	DA2	DA3	DA4	DA5
2	4	2	2	3	4	3	4	2	2	2	3	4	4	2	4	5
5	5	4	3	5	5	5	3	4	5	5	5	5	4	3	5	5
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
3	4	3	3	5	4	4	4	4	2	4	4	4	4	2	4	4
5	5	4	5	5	5	5	5	5	3	3	4	5	3	3	5	5
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
4	4	4	4	4	4	4	4	4	3	4	2	4	3	2	3	3
5	5	4	5	4	5	5	5	4	5	5	5	5	5	4	5	5
4	4	4	2	4	5	4	4	5	3	4	3	4	4	4	4	4
5	5	5	5	4	4	4	4	4	4	5	4	5	4	5	4	5
5	5	4	5	5	5	5	5	5	5	5	5	5	5	5	5	5
4	4	5	4	4	5	3	3	3	3	3	3	5	4	4	5	5
4	3	4	5	5	5	5	5	4	2	3	5	4	5	3	5	5
4	4	5	3	4	5	4	5	4	3	4	4	5	4	4	4	5
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
4	4	4	4	4	4	4	4	4	4	4	4	4	4	5	5	5
5	2	3	4	2	4	5	5	5	2	5	5	4	3	3	5	5
4	4	3	4	3	4	4	3	3	2	3	3	3	3	3	3	4
4	3	3	4	5	5	5	3	4	3	4	5	5	3	2	4	5
5	5	1	2	5	5	5	3	5	1	4	4	5	3	3	4	5
3	2	3	3	3	5	5	5	5	5	5	5	4	2	4	5	5
3	3	3	3	3	5	4	5	5	3	3	3	4	2	3	4	4
4	4	4	3	5	5	5	4	4	4	4	4	4	4	2	5	4
2	1	2	2	2	2	2	2	2	3	4	3	3	1	3	3	3
4	4	4	4	3	3	3	4	4	3	4	4	5	3	3	4	4
5	5	5	4	5	5	5	5	5	4	5	4	5	3	5	4	5

Raw data collected from questionnaire after cleaning and categorizing in each indicator and variable.

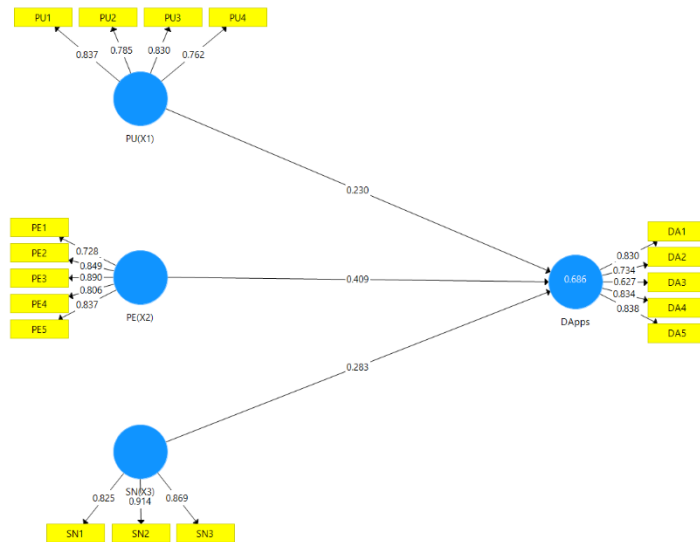
VARIABLE INDICATORS	SCORES					TOTAL	MEAN	PERCENTAGE					
	1	2	3	4	5			1	2	3	4	5	TOTAL
PU1	2	8	50	116	72	248	4.00	0.81%	3.23%	20.16%	46.77%	29.03%	100%
PU2	3	18	47	106	74	248	3.93	1.21%	7.26%	18.95%	42.74%	29.84%	100%
PU3	3	6	79	100	60	248	3.84	1.21%	2.42%	31.85%	40.32%	24.19%	100%
PU4	1	16	72	99	60	248	3.81	0.40%	6.45%	29.03%	39.92%	24.19%	100%
TOTAL	9	48	248	421	266	992	3.90	0.91%	4.84%	25.00%	42.44%	26.81%	100%

VARIABLE INDICATORS	SCORES					TOTAL	MEAN	PERCENTAGE					
	1	2	3	4	5			1	2	3	4	5	TOTAL
PE1	3	22	76	92	55	248	3.70	1.21%	8.87%	30.65%	37.10%	22.18%	100%
PE2	1	3	27	108	109	248	4.29	0.40%	1.21%	10.89%	43.55%	43.95%	100%
PE3	1	6	42	107	92	248	4.14	0.40%	2.42%	16.94%	43.15%	37.10%	100%
PE4	1	6	45	115	81	248	4.08	0.40%	2.42%	18.15%	46.37%	32.66%	100%
PE5	2	7	34	118	87	248	4.13	0.81%	2.82%	13.71%	47.58%	35.08%	100%
TOTAL	8	44	224	540	424	1240	4.07	0.65%	3.55%	18.06%	43.55%	34.19%	100%

VARIABLE INDICATORS	SCORES					TOTAL	MEAN	PERCENTAGE					
	1	2	3	4	5			1	2	3	4	5	TOTAL
SN1	5	19	99	77	48	248	3.58	2.02%	7.66%	39.92%	31.05%	19.35%	100%
SN2	3	11	67	105	62	248	3.85	1.21%	4.44%	27.02%	42.34%	25.00%	100%
SN3	2	8	56	111	71	248	3.97	0.81%	3.23%	22.58%	44.76%	28.63%	100%
TOTAL	10	38	222	293	181	744	3.80	1.34%	5.11%	29.84%	39.38%	24.33%	100%

VARIABLE INDICATORS	SCORES					TOTAL	MEAN	PERCENTAGE					
	1	2	3	4	5			1	2	3	4	5	TOTAL
DA1	2	4	32	117	93	248	4.19	0.81%	1.61%	12.90%	47.18%	37.50%	100%
DA2	4	24	88	82	50	248	3.60	1.61%	9.68%	35.48%	33.06%	20.16%	100%
DA3	4	29	108	69	38	248	3.44	1.61%	11.69%	43.55%	27.82%	15.32%	100%
DA4	1	7	55	109	76	248	4.02	0.40%	2.82%	22.18%	43.95%	30.65%	100%
DA5	1	6	42	109	90	248	4.13	0.40%	2.42%	16.94%	43.95%	36.29%	100%
TOTAL	12	70	325	486	347	1240	3.88	0.97%	5.65%	26.21%	39.19%	27.98%	100%

Notes: The Elements of Perceived Usefulness (Code: PU1, PU2, PU3, PU4). The Elements of Perceived Ease of Use (Code: PE1, PE2, PE3, PE4, PE5). The Elements of Subjective Norm (Code: SN1, SN2, SN3). The Elements of Digital Apps Acceptance (Code: DA1, DA2, DA3, DA4, DA5). Data is processed and analyzed based on the results of the questionnaire answers to be used for further data testing.



The PLS analysis consist of outer model test and inner model test. In the outer model, there is an analysis of the validity and reliability of the variable indicators. Indicator validity consists of convergent validity and discriminant validity, while reliability can be seen from the composite reliability values. The validity test performed to check that two different statements or questions should not produce the same answers, so that there is no data bias (Hair et al., 2019).

Calculation Results of Initial Model Outer Loadings

Source: Data Processed by Smart PLS 3.0, 2023

Convergent validity is assessed by examining the outer loadings table. The loading factor threshold is set at 0.5. If the loading factor value is > 0.5, then convergent validity is satisfied, while if the loading factor value is < 0.5, the construct should be dropped from the analysis (Hair et al., 2019: 112-113).

Outer Loadings

	EA(Y)	PE(X2)	PU(X1)	SN(X3)
EA1	0.830			
EA2	0.734			
EA3	0.627			
EA4	0.834			
EA5	0.838			
PE1		0.728		
PE2		0.849		
PE3		0.890		
PE4		0.806		
PE5		0.837		
PU1			0.837	
PU2			0.785	
PU3			0.830	
PU4			0.762	
SN1				0.825
SN2				0.914
SN3				0.869

Results of Convergent Validity Value Measurement

VARIABLE	INDICATOR	OUTER LOADING VALUE	RESULT
Perceived Usefulness (X1)	PU1	0.837	Valid
	PU2	0.785	Valid
	PU3	0.830	Valid
	PU4	0.762	Valid
Perceived Ease of Use (X2)	PE1	0.728	Valid
	PE2	0.849	Valid
	PE3	0.890	Valid
	PE4	0.806	Valid
	PE5	0.837	Valid
Subjective Norm (X3)	SN1	0.825	Valid
	SN2	0.914	Valid
	SN3	0.869	Valid
Digital Apps Acceptance (Y)	DA1	0.830	Valid
	DA2	0.734	Valid
	DA3	0.627	Valid
	DA4	0.834	Valid
	DA5	0.838	Valid

Source: Data Processed by Smart PLS 3.0

Based on Table, it can be seen that proxies with outer loading factor values greater than 0.5 are assumed to be feasible as indicators that can reflect their respective corresponding variables.

The discriminant validity of the indicators using SmartPLS version 3.0 is checked based on three criteria called Cross Loading criterion, Fornell and Larcker's criterion and Heterotrait-Monotrait ratio of correlations (HTMT) (Hair et al., 2019: 114-115).

Cross Loading Test

Cross Loading test involves comparing the correlations of an indicator between construct. The indicators should have higher loadings on their intended construct than on other constructs.

Discriminant Validity

Fornell-Larcker Criterion

Cross Loadings

Heterotrait-Monotrait Ratio (HTMT)

	EA(Y)	PE(X2)	PU(X1)	SN(X3)
EA1	0.830	0.757	0.648	0.634
EA2	0.734	0.532	0.548	0.532
EA3	0.627	0.408	0.464	0.420
EA4	0.834	0.623	0.527	0.611
EA5	0.838	0.646	0.527	0.600
PE1	0.532	0.728	0.496	0.493
PE2	0.649	0.849	0.569	0.555
PE3	0.697	0.890	0.649	0.649
PE4	0.683	0.806	0.575	0.619
PE5	0.634	0.837	0.639	0.677
PU1	0.671	0.641	0.837	0.599
PU2	0.507	0.541	0.785	0.436
PU3	0.545	0.554	0.830	0.512
PU4	0.511	0.547	0.762	0.486
SN1	0.497	0.516	0.503	0.825
SN2	0.680	0.668	0.572	0.914
SN3	0.693	0.694	0.584	0.869

Results of Discriminant Validity Value Measurement

IND	PU(X1)	PE(X2)	SN(X3)	DApps(Y)
PU1	0.837	0.641	0.599	0.671
PU2	0.785	0.541	0.436	0.507
PU3	0.830	0.554	0.512	0.545
PU4	0.762	0.547	0.486	0.511
PE1	0.496	0.728	0.493	0.532
PE2	0.569	0.849	0.555	0.649
PE3	0.649	0.890	0.649	0.697
PE4	0.575	0.806	0.619	0.683
PE5	0.639	0.837	0.677	0.634
SN1	0.503	0.516	0.825	0.497
SN2	0.572	0.668	0.914	0.680
SN3	0.584	0.694	0.869	0.693
DA1	0.648	0.757	0.634	0.830
DA2	0.548	0.532	0.532	0.734
DA3	0.464	0.408	0.420	0.627
DA4	0.527	0.623	0.611	0.834
DA5	0.527	0.646	0.600	0.838

Source: Data Processed by Smart PLS 3.0

Based on Table, it can be seen that the cross-loading values for each indicator of each latent variable are higher than the cross-loading values when connected to other latent variables. This means that each latent variable already has good discriminant validity, where latent variables don't have high correlations and similarities with other constructs.

Fornell and Larcker's (1981) Test

The value of Fornell and Larcker's (1981) test is the root of average variance extracted (AVE). It is suggested that the square root of AVE of each latent variable should be greater than the correlations among the latent variables.

Discriminant Validity

Fornell-Larcker Criterion	Cross Loadings	Heterotrait-Monotrait Ratio (HTMT)			
	EA(Y)	PE(X2)	PU(X1)	SN(X3)	
EA(Y)	0.777				
PE(X2)	0.780	0.824			
PU(X1)	0.703	0.714	0.804		
SN(X3)	0.728	0.730	0.639	0.870	

Results of Fornell and Larcker's Measurement (Step 1)

VAR	PU(X1)	PE(X2)	SN(X3)	DApps(Y)
PU(X1)	0.804			
PE(X2)	0.714	0.824		
SN(X3)	0.639	0.730	0.870	
DApps(Y)	0.703	0.780	0.728	0.777

Source: Data Processed by Smart PLS 3.0

Based on the results of the Fornell and Larcker's test in Table above, it is shown that the square root of the Average Variance Extracted (AVE) of each latent variable is greater than the square root of the AVE of other latent variables, except for the relationship between Digital Apps Acceptance (Y) and Perceived Ease of Use (X2). This indicates that the latent variables have not

demonstrated good discriminant validity, meaning that the indicators within the construct block are not better than the indicators in other blocks.

Based on the results of Fornell and Larcker's test, the indicator DA1 from the Digital Apps Acceptance (Y) variable was removed from the model as it was found to have high correlation with the Perceived Ease of Use (X2) variable based on indicator correlations.

	PE1	PE2	PE3	PE4	PE5	
DA1	0.54	0.632	0.695	0.593	0.651	0.6222
DA2	0.355	0.423	0.485	0.505	0.408	0.4352
DA3	0.295	0.291	0.354	0.447	0.284	0.3342
DA4	0.419	0.556	0.536	0.533	0.511	0.511
DA5	0.416	0.551	0.579	0.561	0.536	0.5286
	0.405	0.4906	0.5298	0.5278	0.478	

Correlation test between Digital Apps Acceptance (Y) with the Perceived Ease of Use (X2) indicators which found DA1 to have the highest correlation between two variables.

After dropping the indicator, the second stage of the Fornell and Larcker's test for discriminant validity was conducted and the output is presented below:

Discriminant Validity

Fornell-Larcker Criterion	Cross Loadings	Heterotrait-Monotrait Ratio (HTMT)			
		DApps	PE(X2)	PU(X1)	SN(X3)
DApps		0.784			
PE(X2)		0.716	0.824		
PU(X1)		0.660	0.714	0.804	
SN(X3)		0.697	0.730	0.640	0.870

Results of Fornell and Larcker's Measurement (Step 2)

VAR	PU(X1)	PE(X2)	SN(X3)	DApps(Y)
PU(X1)	0.804			
PE(X2)	0.714	0.824		
SN(X3)	0.640	0.730	0.870	
DApps(Y)	0.660	0.716	0.697	0.784

Source: Data Processed by Smart PLS 3.0

Table above indicates that all individual constructs exhibit sufficient or satisfactory discriminant validity using Fornell and Larcker (1981) discriminant validity assessment, where the square root of AVE is larger than the correlations for all other reflective constructs.

Heterotrait-Monotrait Ratio of Correlations (HTMT)

HTMT refers to the ration of correlations of indicators towards their own construct against the correlations of the indicators towards other constructs (Ramayah et.al., 2016). HTMT estimates the actual correlation between two constructs assuming that the constructs are perfectly measured and reliable with free of error. There are two methods to measure discriminant validity using HTMT. When using the measurement as a criterion, it can be evaluate using either HTMT value greater than HTMT.85 value of 0.85 (Kline, 2011) or HTMT.90 value of 0.90 (Gold, Malhotra and Segars, 2001) to indicate the discriminant validity.

Discriminant Validity

Fornell-Larcker Criterion

Cross Loadings

Heterotrait-Monotrait Ratio (HTMT)

	DApps	PE(X2)	PU(X1)	SN(X3)
DApps				
PE(X2)	0.842			
PU(X1)	0.810	0.834		
SN(X3)	0.836	0.833	0.758	

As presented in Table above, all the values carried by each construct tested fulfil the criterion of $HTMT < 0.85$ (Kline, 2011). Therefore, this indicated that discriminant validity has been ascertained. Besides, the result of HTMT inference based on the confidence interval value also shows that the confidence interval does not show a value of 1 on any of the constructs (Henseler et.al., 2015), which also confirmed discriminant validity.

Reliability Test

Reliability testing is necessary to measure the stability and consistency of an instrument in measuring a concept or variable. Reliability criteria can also be seen from the reliability value of a construct (Hair et al., 2019: 111-112). To determine whether a measuring instrument is reliable or not, a reliability coefficient is calculated. The reliability coefficient should be greater than 0.70.

Construct Reliability and Validity

	Cronbach's Alpha	rho_A	Composite Reliability	Average Variance Extracted (AVE)
DApps	0.789	0.805	0.864	0.615
PE(X2)	0.881	0.889	0.913	0.679
PU(X1)	0.818	0.835	0.879	0.646
SN(X3)	0.841	0.860	0.903	0.757

The reliability metrics show consistently robust internal consistency, with Cronbach's Alpha values ranging from 0.789 to 0.881, all exceeding the recommended threshold of 0.7. The more precise Rho_A reliability indicator further confirms this consistency with values between 0.805 and 0.889. Composite Reliability values are particularly strong, ranging from 0.864 to 0.913, indicating excellent overall construct reliability. The Average Variance Extracted (AVE), is similarly strong across all constructs with values from 0.615 to 0.757, well above the 0.5 threshold. This indicates that each construct explains a substantial portion of the variance in its indicators.

Performance Expectancy (PE/X2) demonstrates the highest overall reliability with a Composite Reliability of 0.913, while Subjective Norm (SN/X3) shows the strongest convergent validity with an AVE of 0.757. These results collectively suggest a robust measurement model with constructs that are both internally consistent and well-differentiated, providing a solid foundation for subsequent structural model assessment and hypothesis testing. The DApps construct, while showing acceptable reliability ($\alpha=0.789$), has the lowest scores across all metrics, suggesting potential for refinement in future research, though its current values remain within acceptable ranges for all reliability and validity criteria.

Structural model testing or inner model testing is conducted to measure the overall relationship between variables in this study. Inner model measurement is carried out to determine the level of influence of the relationship between variables, as well as the level of influence of the overall relationship between variables in the system that has been

constructed. Inner model measurement to test the influence between variables in the study for the reflective indicators model uses the value of R² (Coefficient of Determination), Q² (Predictive Relevance) and F² (Effect Size) (Hair et al., 2019: 76-77).

Coefficient of Determination R Square (R²)

R Square (R²), also known as the coefficient of determination, is a measure of the goodness of fit of a regression equation. It provides the proportion or percentage of the total variation in the dependent variable that is explained by the independent variables. The value of R² ranges from 0 to 1, and a better fit is indicated by a value that approaches 1 (Hair et al., 2019: 114-115). The R² value of a model can be classified as strong (≤ 0.67), moderate (≤ 0.33), and weak (≤ 0.19). The following are the results of R² values in the research model:

Quality Criteria

R Square

	R Square	R Square Adjusted
DApps	0.686	0.683

Based on Table, the R² value indicates that the R² value of Digital Apps Acceptance (Y) is 0.686, which means that the variables Perceived Usefulness (X1), Perceived Ease of Use (X2), and Subjective Norm (X3) can explain the variation of Digital Apps Acceptance (Y) up to 68.6%.

Predictive Relevance Q Square (Q²)

According to Hair et al. (2019: 127-131), predictive relevance or Q² is a measure of the ability of a model to predict future observations based on the data used to estimate the model. It measures the difference between the actual values and the predicted values of the dependent variable, adjusted for the number of predictor variables and the degrees of freedom. Q² is an important measure of model validity because it assesses the model's ability to make accurate predictions for new observations beyond the original data set used for model estimation. A Q² value greater than zero indicates that the model has predictive relevance, meaning that it can be used to make accurate predictions. A Q² value of 0.25 or higher is considered to be strong evidence of predictive relevance.

Construct Crossvalidated Redundancy

Total	Case1	Case2	Case3	Case4	Case5	Case6	Case7
	SSO			SSE			Q ² (=1-SSE/SSO)
DApps	992.000			636.189			0.359
PE(X2)	1240.000			1240.000			
PU(X1)	992.000			992.000			
SN(X3)	744.000			744.000			

Based on the results of the Q-square analysis, it can be interpreted that the variables Perceived Usefulness, Perceived Ease of Use, and Subjective Norm that influence Digital Apps Acceptance have a predictive relevance (Q²) value of 0.359. This means that the structural model in this study estimates parameters and produces values for each variable of $0.359 > 0.25$, indicating that the model has a strong predictive relevance (Q²) value.

Effect Size F Square (F2)

According to Hair et al. (2019: 127-131), Effect size, also known as the F-square or F2, is a measure used in PLS-SEM to determine the proportion of variance in the endogenous construct that can be explained by the exogenous construct(s). The F2 value is calculated by dividing the R2 value of the endogenous construct by the proportion of variance explained by the exogenous construct(s) that directly or indirectly affect the endogenous construct. An F2 value of 0.02 is considered small, 0.15 is moderate, and 0.35 is large, indicating that the exogenous construct(s) have a small, moderate, or large effect on the endogenous construct, respectively.

f Square

	DApps
DApps	
PE(X2)	0.194
PU(X1)	0.078
SN(X3)	0.112

Based on the results of the F-square test above, it can be interpreted that the Perceived Usefulness variable on Digital Apps Acceptance has a large effect size (F2) with a value of 0.059. Perceived Ease of Use on Digital Apps Acceptance has a large effect size (F2) with a value of 0.095. Subjective Norm on Digital Apps Acceptance has a large effect size (F2) with a value of 0.110. This suggests that these variables are important predictors of Digital Apps Acceptance in the model. Therefore, it can be concluded that these three variables are important factors that significantly influence Digital Apps Acceptance.

Hypothesis Testing

The estimation coefficient value between constructs must have a significant value. The significance of the relationship can be obtained through the Bootstrapping or Jackknifing procedure. The resulting value is in the form of a t-value/t-statistic, which is then compared to the t-table. If the calculated t-value is greater than the t-table value (1.96) at a significance level of 5%, then the estimation coefficient value for the path is significant. In addition, hypothesis testing can also be done by examining the probability value. If the p-value is less than 0.05, then the hypothesis is considered significant or accepted, and if the p-value is greater than 0.05, then the hypothesis is considered not significant or rejected. This is done to strengthen assumptions in hypothesis testing by providing alternative hypothesis testing methods (Hair et al., 2019: 120).

Path Coefficients

Mean, STDEV, T-Values, P-Values	Confidence Intervals	Confidence Intervals Bias Corrected	Samples		
	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values
PE(X2) -> DApps	0.323	0.321	0.074	4.365	0.000
PU(X1) -> DApps	0.227	0.232	0.065	3.484	0.000
SN(X3) -> DApps	0.316	0.315	0.057	5.588	0.000

1. H1: Perceived Usefulness Affects the Students' Digital Apps Acceptance.

Based on the test results in the table above, the influence of Perceived Usefulness on Digital Apps Acceptance showed a positive coefficient value of 0.230, t-Statistic of 4.143 > t-table (1.96), and P Values of 0.000 < 0.05, therefore Hypothesis H1 is accepted. This can be interpreted as the

influence of Perceived Usefulness on Digital Apps Acceptance is proven to have a positive and significant effect.

2. H2: Perceived Ease of Use Affects the Students' Digital Apps Acceptance.

Based on the test results in the table above, the influence of Perceived Ease of Use on Digital Apps Acceptance showed a positive coefficient value of 0.409, t-Statistic of $6.367 > t\text{-table (1.96)}$, and P Values $0.000 < 0.05$, therefore Hypothesis H2 is accepted. This can be interpreted as the influence of Perceived Ease of Use on Digital Apps Acceptance is proven to be positive and significant.

3. H3: Subjective Norm Affects the Students' Digital Apps Acceptance.

Based on the test results in the table above, the influence of Subjective Norm on Digital Apps Acceptance showed a positive coefficient value of 0.283, a t-Statistic value of $5.480 > t\text{-table (1.96)}$, and P Values of $0.000 < 0.05$, therefore Hypothesis H3 is accepted. This can be interpreted as the influence of Subjective Norm on Digital Apps Acceptance is proven to be positive and significant.

Conclusion

Based on the findings from the statistical analysis conducted in this study, it can be concluded that the three hypotheses proposed have been supported with positive and significant influence towards students' Digital Apps Acceptance.

Firstly, the results indicate that the Perceived Usefulness of Digital Apps is positively associated with students' Digital Apps Acceptance. This means that students who perceive Digital Apps as useful are more likely to accept and use it. The significant impact of Perceived Usefulness highlights the importance of ensuring that Digital Apps are designed to meet the needs of students, providing them with clear and tangible benefits.

Secondly, the Perceived Ease of Use of Digital Apps was also found to have a positive and significant influence on students' acceptance. This suggests that students are more likely to use Digital Apps if they are easy to use and understand. This highlights the importance of developing Digital Apps with a user-friendly interface, providing students with easy access to their accounts, and clear and concise instructions on how to use them.

Lastly, the results indicate that Subjective Norm also has a positive and significant effect on students' Digital Apps Acceptance. This implies that students' perception of the social influence and pressure from their peers, family, or significant others play a role in shaping their acceptance of Digital Apps. Therefore, it is important for Digital Apps providers to understand the social factors that influence students' acceptance and design marketing strategies that appeal to these factors.

Project 4

Analysis of the Relationship between Industrial Efficiency and Export Performance, Mediated by Competitive Strategy, in Processed Cocoa Agroindustry in Indonesia.

A project example featuring a more complex model and measurement approach.

X1	X2	X3	X4	X5	X6	X7	X8	Z1.1	Z1.2	Z1.3	Z1.4	Z2.1	Z2.2	Z2.3	Z3.1	Z3.2	Z3.3	Z3.4	Z4.1	Z4.2	Z4.3	Z4.4
62400	17500	250000	60000	1900	180000	127800	52200	4.833333	4.944444	3.583333	4.5	4.777778	3.222222	4.75	4.4	4.8	4.666667	4.8	4.433333	4.066667	4.4	4.533333
46800	12600	200000	30000	1100	135000	81000	54000	4.777778	4.777778	3.5	4.416667	4.666667	3.055556	4.75	4.3	4.633333	4.583333	4.8	4.333333	4.133333	4.066667	4.3
58968	10500	200000	10000	970	162000	115020	46980	4.722222	4.888889	3.583333	4.333333	4.666667	2.833333	4.583333	4.4	4.666667	4.472222	4.733333	4.2	4.2	4.333333	4.333333
31200	9500	77000	20000	630	90000	54000	36000	4.944444	4.777778	3.333333	4.416667	4.944444	3.055556	4.833333	4.333333	4.7	4.5	4.8	3.611111	3.444444	4.133333	4.433333
46800	8500	90000	25000	1224	135000	94500	40500	4.944444	4.944444	3.666667	4.416667	5	2.722222	4.916667	4.333333	4.333333	4.638889	4.8	4.4	4	4.233333	4.466667
31200	8500	83000	20000	960	133000	86450	46550	4.833333	4.888889	3.583333	4.25	4.777778	3.055556	4.791667	4.266667	4.7	4.388889	4.766667	4.333333	4.066667	4.166667	4.266667
46800	9500	98000	30000	1405	180000	126000	54000	4.833333	4.611111	3.666667	4.333333	4.833333	3.166667	5	4.333333	4.533333	4.333333	4.8	4	3.933333	4.233333	4.333333
31200	6500	55000	20000	433	90000	63000	27000	4.888889	4.666667	3.333333	4.5	4.833333	3.611111	4.875	4.366667	4.666667	4.527778	4.833333	4.3	4.066667	4.166667	4.2
46800	8500	99000	30000	1402	155250	87750	67500	4.833333	4.888889	3.333333	4.5	4.722222	3.5	4.791667	4.4	4.766667	4.388889	4.8	4.266667	4	3.866667	4.433333
46800	6500	85000	35000	1608	214650	119250	95400	4.777778	4.944444	3.583333	4.5	4.888889	2.722222	4.833333	4.4	4.533333	4.361111	4.8	3.305556	3.222222	4.366667	4.433333
31200	4500	75000	20000	522	85500	54000	31500	4.833333	4.833333	3.5	4.5	5	2.666667	4.916667	4.266667	4.766667	4.472222	4.6	4.233333	4.2	4.133333	4.133333
46800	7512	98000	25000	649	105750	58500	47250	4.888889	4.944444	3.666667	4.5	5	2.611111	5	4.366667	4.733333	4.5	4.566667	4.566667	4.066667	3.966667	4.3
38000	8592	85000	30000	369	90000	54000	36000	5	4.777778	3.583333	4.416667	4.777778	2.611111	4.833333	4.4	4.733333	4.611111	4.666667	3.555556	3.888889	4.366667	4.533333
31500	4534	80000	20000	531	90000	54000	36000	4.722222	4.777778	3.5	4.5	4.944444	2.555556	4.916667	4.4	4.533333	4.527778	4.633333	4.366667	4.066667	4.333333	4.466667
33000	6555	95000	17500	1600	178800	107280	71520	4.777778	4.833333	3.5	4.5	4.777778	2.611111	4.625	4.366667	4.733333	4.305556	4.533333	4.433333	4	4.333333	4.5
30500	4711	90000	10000	552	90000	54000	36000	5	4.666667	3.666667	4.416667	4.888889	2.666667	4.625	4.3	4.733333	4.583333	4.433333	3.694444	3.888889	4.4	4.433333
23400	5622	50000	10000	335	67500	43875	23625	4.888889	4.777778	3.583333	4.5	4.722222	2.722222	4.833333	4.366667	4.566667	4.472222	4.666667	4.4	4.133333	4.333333	4.133333
15600	6588	40000	10000	301	45000	22500	22500	4.944444	4.833333	3.5	2.083333	5	2.722222	4.958333	4.466667	4.666667	4.277778	4.566667	4.466667	4.033333	4.3	4.433333
19344	5950	60000	20000	1211	73800	40590	33210	4.777778	5	3.583333	4.5	4.888889	2.666667	5	4.266667	4.366667	4.416667	4.8	3.416667	3.777778	4.2	4.533333
23400	6500	70000	20000	300	67500	33750	33750	4.888889	4.944444	3.5	4.5	5	2.666667	4.75	4.366667	4.4	4.638889	4.566667	4.3	3.966667	4.3	3.966667
15500	6500	70000	15000	355	45000	24750	20250	4.833333	4.722222	3.5	4.416667	5	2.666667	4.875	4.266667	4.633333	4.472222	4.633333	4.4	4.066667	4.166667	4.366667
14300	7500	70000	10000	224	31500	15750	15750	4.944444	4.888889	3.416667	4.5	5	2.555556	4.791667	4.333333	4.633333	4.555556	4.733333	3.472222	3.416667	4.166667	4.333333
31200	2510	50000	8000	723	12720	6360	6360	4.833333	4.833333	3.5	4.5	4.722222	2.666667	4.875	4.333333	4.8	4.527778	4.433333	4.4	4.1	4	4.166667
30100	3500	40000	6000	531	9000	5400	3600	4.833333	4.666667	3.5	4.5	5	2.833333	4.958333	4.3	4.6	4.527778	4.433333	4.3	3.933333	4.066667	4.433333
62400	2820	40000	200	187	1800	200	1800	4.833333	4.555556	3.5	4.166667	4.666667	3.111111	5	4.3	4.3	4.333333	4.5	3.611111	3.361111	4.066667	4.233333
30200	561	1800	150	226	2700	400	2700	4.833333	4.888889	3.5	4.5	4.888889	3.722222	4.875	4.366667	4.6	4.305556	4.6	4.133333	4.033333	4.066667	4.166667
10000	620	110	7.2	1640	16290	4116	1629	4.944444	4.888889	3.416667	4.333333	5	3.222222	4.791667	4.333333	4.8	4.527778	4.4	3.8	3.466667	3.9	3.9

Relationship between Concepts, Variables and Research Indicators

Variable	Dimension	Indicator	Measurement Techniques
(X) Industrial Efficiency	X.1. Raw Materials	X1. Raw Materials (Processed Cocoa Beans)	Questionnaire Company Data Source
	X.2. Labor Costs	X2. Cocoa Agroindustry Labor Costs	
	X.3. Capital	X3. Cocoa Agroindustry Capital	
	X.4. Production Capacity	X4. Factory Production Capacity	
	X.5. Number of Workers	X4. Number of Cocoa Agroindustry Labor	
	X.6. Processed Cocoa Production	X6. Processed Cocoa Products Produced	
	X.7. Export Products	X7. Processed Cocoa Export Products	
	X.8. Local Product Distribution	X8. Distribution of products to the local market	
(Z) Competitive Strategy	Z.1. Business Environment (Cho and Moon, 2000)	Z1.1. Availability of Labor	Questionnaire Likert Scale (Score 1-5)
		Z1.2. Industrial Competition	
		Z1.3. Protection Policy	
		Z1.4. National Political and Economic Conditions	
	Z.2. Endowment Factors (Cho and Moon, 2000)	Z2.1. Cocoa Bean Production	
		Z2.2. Business Actors	
		Z2.3. Differentiation of Processed Cocoa Products	
	Z.3. Supply Chain Food Supply Chain Networking Framework (Van der Vrost, 2006)	Z3.1. Agro-Industry Supply Chain Management	
		Z3.2. Agro-Industry Supply Chain Structure	
		Z3.3. Agro-Industry Supply Chain Resources	
	Z.4. Entrepreneurial Orientation (Lee & Tsang, 2001)	Z3.4. Agro-Industry Supply Chain Process	
		Z4.1. Need For Achievement	
		Z4.2. Self-Reliance	
		Z4.3. Extroversion	
		Z4.4. Internal Locus of Control	
(Y) Export Performance	Y.1. Management Characteristics and Perceptions	Y1.1. Support and commitment	Questionnaire Likert Scale (Score 1-5)
		Y1.2. Export Experience	
		Y1.3. International Orientation	
		Y1.4. Motivation for Cocoa Agroindustry Export	
		Y1.5. Perception of Benefits	
	Y.2. Export Strategy	Y2.1. General Export Strategy	
		Y2.2. Export Planning	
		Y2.3. Proactive	

Y.3. Export Knowledge

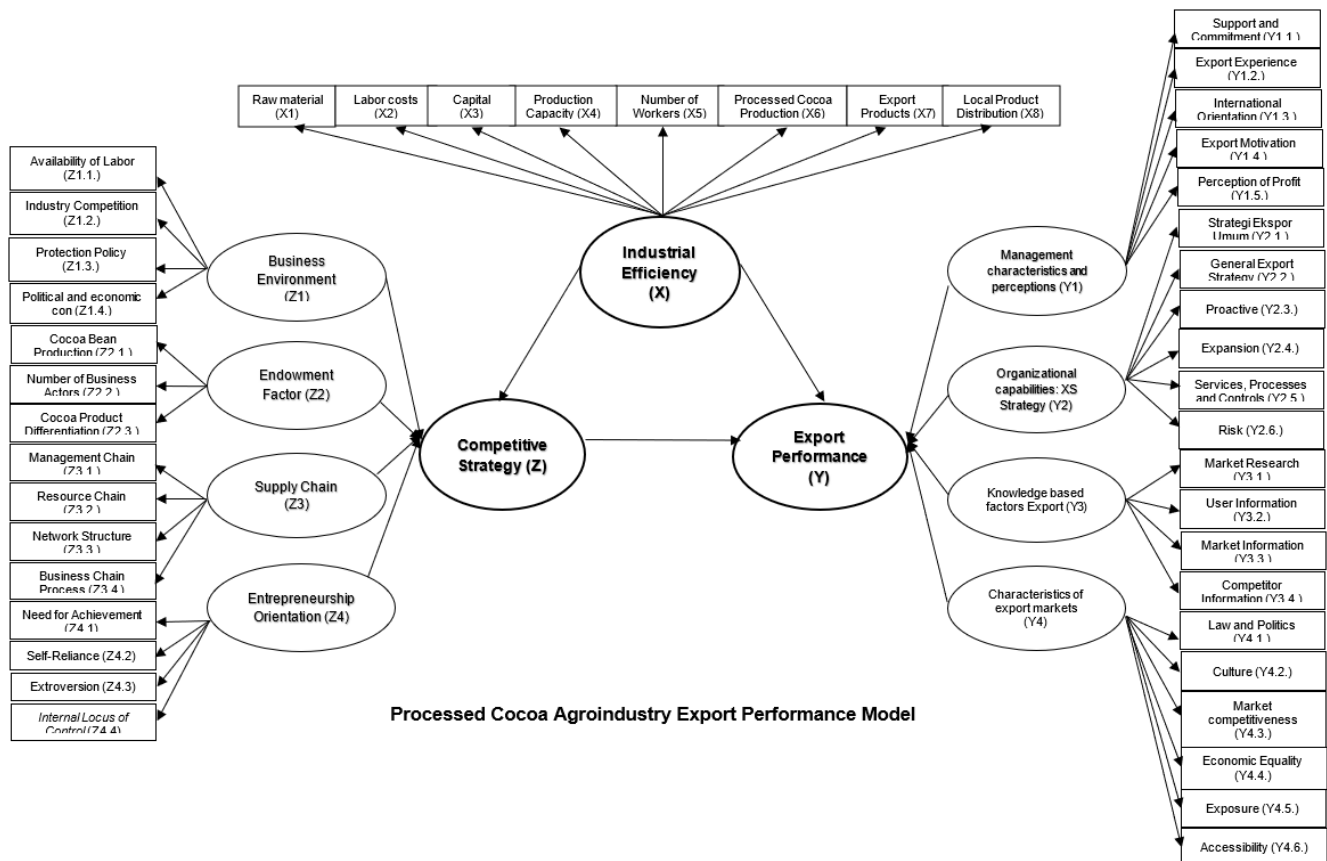
Y.4. Export Market Characteristics

- Y2.4. Expansion
- Y2.5. Services, Processes and Control
- Y2.6. Risk
- Y3.1. Market Research
- Y3.2. User Information
- Y3.3. Market Information
- Y3.4. Competitor Information
- Y4.1. Law and Politics
- Y4.2. Culture
- Y4.3. Market Competitiveness
- Y4.4. Economic Equality (Zoning areas for export: Asia, America, Europe)
- Y4.5. Exposure (Product promotion through websites, emails and e-commerce)
- Y4.6. Accessibility

Source: (Goldin, 1990; Cho and Moon, 2000; Lee & Tsang, 2001; Van der Vrost, 2006; Abdoellah, 2021)

Raw data and each variable indicators measurement method.

* data obtained from financial reports of Processed Cocoa Agroindustry in Indonesia for the period 2022 (Central Bureau of Statistics, Plantation Company Survey) and structured self-administered questionnaire survey that was adapted from past investigations and analysis (Goldin, 1990; Cho and Moon, 2000; Lee & Tsang, 2001; Van der Vrost, 2006; Abdoellah, 2021).



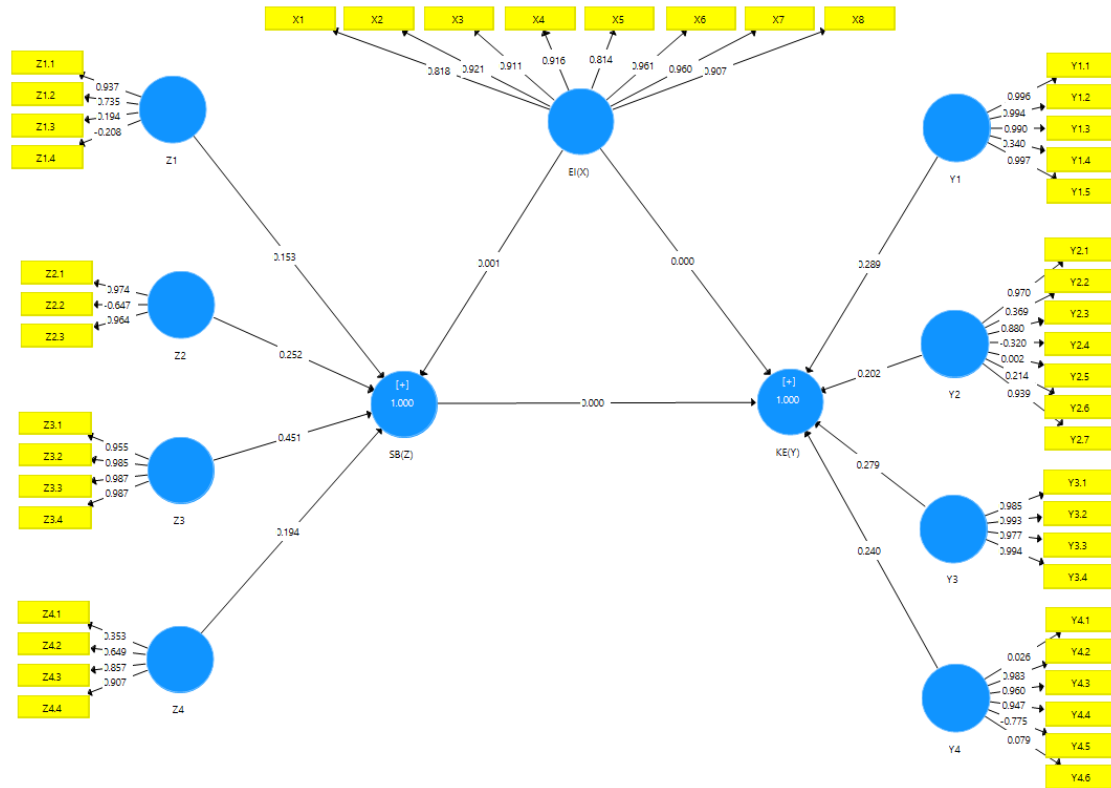


Image of original research model and PLS model based on

The image presents a comprehensive structural model for analyzing export performance in the Indonesian processed cocoa agroindustry. Breakdown of the key relationships:

1. Central Construct: Industrial Efficiency (X)

- This is the core variable that captures the overall operational capabilities of the cocoa agroindustry
- Influenced by multiple input factors like raw materials, labor costs, capital, production capacity, and number of workers
- Shaped by external environment factors such as business environment, labor availability, industry competition, and policy conditions

2. Mediating Variable: Competitive Strategy (Z)

- Acts as a critical intermediary between Industrial Efficiency and Export Performance
- Incorporates elements like:
 - Management characteristics
 - Organizational capabilities
 - Entrepreneurship orientation
 - Supply chain management
 - Network structure
 - Self-reliance
 - Need for achievement

3. Outcome Variable: Export Performance (Y)

- **The ultimate dependent variable representing the success of cocoa exports**
- **Influenced directly by Industrial Efficiency and Competitive Strategy**
- **Includes sub-factors like:**
 - **Export motivation**
 - **Perception of profit**
 - **Strategic export intention**
 - **Characteristics of export markets**
 - **Knowledge-based export factors**

The model suggests a complex, interdependent relationship where:

- **Industrial efficiency is not just about operational metrics**
- **Competitive strategy transforms industrial capabilities into export success**
- **Multiple internal and external factors interact to determine export performance**

The key insight is that export performance is not a direct result of industrial efficiency, but is mediated by how effectively a firm can develop and implement a competitive strategy that leverages its industrial capabilities.

Industrial Efficiency does not operate in isolation but is deeply embedded in a complex ecosystem of internal and external influences. The business environment, industry competition, protection policies, and political-economic conditions all shape the foundational capabilities of cocoa agroindustries. These contextual factors interact with more granular elements like the number of workers, processed cocoa production levels, and the inherent endowment factors of the industry.

Export Performance emerges as the ultimate outcome of this intricate process. It is the result of how effectively an agroindustry can convert its industrial efficiency into a compelling competitive strategy. This is not merely about producing cocoa, but about understanding export markets, developing strategic export intentions, perceiving potential profits, and creating knowledge-based export approaches. The model recognizes that successful exporting requires more than just production capacity – it demands a holistic understanding of global market dynamics, strategic adaptation, and continuous innovation.

For Indonesian cocoa agroindustries, this model offers a comprehensive roadmap. It suggests that to excel in international markets, businesses must look beyond traditional metrics of production. They must develop a nuanced approach that combines operational excellence with strategic thinking, market intelligence, and adaptive capabilities. The path to successful cocoa exports lies not just in what is produced, but in how effectively and intelligently that production is strategically positioned and marketed to the global market.

Outer Loading Value Calculation Results

Variable/Dimension		Indicator	Outer Loading Value		Information
			Original Model	Modified Model	
Industrial Efficiency (X)		X1	0.818	0.818	Valid
		X2	0.921	0.921	Valid
		X3	0.911	0.911	Valid
		X4	0.916	0.916	Valid
		X5	0.814	0.815	Valid
		X6	0.961	0.961	Valid
		X7	0.960	0.960	Valid
		X8	0.907	0.908	Valid
Export Performance (Y)	Management Characteristics and Perceptions (Y1)	Y1.1	0.996	0.998	Valid
		Y1.2	0.994	0.996	Valid
		Y1.3	0.990	0.996	Valid
		Y1.4	0.340	-	Drop
		Y1.5	0.997	0.998	Valid
	Export Strategy (Y2)	Y2.1	0.970	0.986	Valid
		Y2.2	0.369	-	Drop
		Y2.3	0.880	0.913	Valid
		Y2.4	-0.320	-	Drop
		Y2.5	0.002	-	Drop
		Y2.6	0.214	-	Drop
		Y2.7	0.939	0.923	Valid
	Export Knowledge (Y3)	Y3.1	0.985	0.985	Valid
		Y3.2	0.993	0.993	Valid
		Y3.3	0.977	0.977	Valid
		Y3.4	0.994	0.994	Valid
	Export Market Characteristics (Y4)	Y4.1	0.026	-	Drop
		Y4.2	0.983	0.995	Valid
		Y4.3	0.960	0.966	Valid
		Y4.4	0.947	0.962	Valid
		Y4.5	-0.775	-	Drop
		Y4.6	0.079	-	Drop
Competitive Strategy (Z)	Business Environment (Z1)	Z1.1	0.937	0.943	Valid
		Z1.2	0.735	0.753	Valid
		Z1.3	0.194	-	Drop
		Z1.4	-0.208	-	Drop
	Endowment Factors (Z2)	Z2.1	0.974	0.994	Valid
		Z2.2	-0.647	-	Drop
		Z2.3	0.964	0.955	Valid

Variable/Dimension	Indicator	Outer Loading Value		Information
		Original Model	Modified Model	
Supply Chain (Z3) Entrepreneurial Orientation (Z4)	Z3.1	0.955	0.985	Valid
	Z3.2	0.985	0.987	Valid
	Z3.3	0.987	0.988	Valid
	Z3.4	0.987	0.955	Valid
	Z4.1	0.353	-	Drop
	Z4.2	0.649	-	Drop
	Z4.3	0.857	0.923	Valid
	Z4.4	0.907	0.949	Valid

The outer loading factor value shows that the proxy value of the Management Characteristics and Perception dimension (Y1) with indicator Y1.4, Organizational Capabilities: Export Strategy (Y2) with indicators Y2.2, Y2.4, Y2.5, Y2.6, Export Market Characteristics (Y4) with indicators Y4.1, Y4.5 and Y4.6 is less than 0.5 and is not suitable for use as an indicator to reflect the Export Performance variable (Y) so it must be removed from the model.

The outer loading factor value shows that the proxy value of the Business Environment dimension (Z1) with indicators Z1.2, Z1.3, Z1.4, Endowment Factor (Z2) with indicator Z2.2, Entrepreneurial Orientation (Z4) with indicators Z4.1 and Z4.2 is less than 0.5 and is not suitable for use as an indicator to reflect the Competitive Strategy variable (Z) so it must be removed from the model.

Based on the Table, it can be seen that the remaining proxies have an outer loading factor value greater than 0.5. So the outer loading factor value is assumed to be feasible to be used as an indicator that can reflect each corresponding variable.

After conducting a convergent validity test through the outer loading value, the next stage is to conduct a discriminant validity test. Discriminant validity can be measured from the cross-loading value with its construct, each indicator in a construct has a difference with the indicators in other constructs which can be indicated by a higher loading value than its own construct (Hair et al., 2019: 114-115).

Cross-loading Value Calculation Results

Indicator	Industrial Efficiency (X)	Export Performance (Y)				Competitive Strategy (Z)			
		Y1	Y2	Y3	Y4	Z1	Z2	Z3	Z4
X1	0.818	0.732	0.727	0.725	0.706	0.582	0.669	0.71	0.61
X2	0.921	0.686	0.729	0.71	0.697	0.637	0.636	0.732	0.744
X3	0.91	0.602	0.653	0.633	0.616	0.548	0.531	0.652	0.684
X4	0.916	0.597	0.67	0.625	0.604	0.568	0.569	0.643	0.692
X5	0.815	0.567	0.607	0.593	0.587	0.576	0.518	0.584	0.498
X6	0.961	0.61	0.652	0.634	0.621	0.578	0.548	0.644	0.694

X7	0.96	0.584	0.625	0.608	0.598	0.553	0.52	0.621	0.68
X8	0.908	0.59	0.637	0.614	0.596	0.567	0.534	0.623	0.678
Y1.1	0.689	0.998	0.964	0.979	0.973	0.871	0.969	0.97	0.772
Y1.2	0.686	0.996	0.984	0.987	0.982	0.889	0.972	0.981	0.782
Y1.3	0.701	0.996	0.971	0.98	0.974	0.874	0.969	0.974	0.806
Y1.5	0.686	0.998	0.972	0.984	0.979	0.882	0.972	0.976	0.79
Y2.1	0.699	0.978	0.986	0.983	0.98	0.876	0.959	0.991	0.807
Y2.3	0.647	0.84	0.913	0.832	0.832	0.757	0.817	0.871	0.734
Y2.7	0.738	0.93	0.923	0.946	0.931	0.887	0.929	0.908	0.742
Y3.1	0.689	0.97	0.948	0.985	0.983	0.917	0.979	0.956	0.764
Y3.2	0.689	0.981	0.969	0.993	0.992	0.914	0.979	0.976	0.789
Y3.3	0.746	0.96	0.972	0.977	0.957	0.868	0.935	0.964	0.801
Y3.4	0.707	0.982	0.985	0.994	0.991	0.91	0.974	0.99	0.816
Y4.2	0.687	0.986	0.976	0.995	0.995	0.92	0.98	0.982	0.808
Y4.3	0.671	0.947	0.917	0.963	0.966	0.92	0.962	0.926	0.732
Y4.4	0.69	0.932	0.955	0.946	0.962	0.857	0.922	0.972	0.803
Z1.1	0.652	0.947	0.955	0.977	0.981	0.943	0.968	0.962	0.79
Z1.2	0.388	0.436	0.453	0.461	0.475	0.753	0.463	0.413	0.395
Z2.1	0.626	0.97	0.951	0.974	0.974	0.904	0.994	0.957	0.763
Z2.3	0.629	0.966	0.957	0.972	0.973	0.901	0.994	0.952	0.744
Z3.1	0.685	0.915	0.906	0.911	0.921	0.779	0.885	0.955	0.794
Z3.2	0.702	0.971	0.977	0.984	0.979	0.911	0.958	0.985	0.792
Z3.3	0.702	0.965	0.983	0.97	0.972	0.876	0.954	0.987	0.811
Z3.4	0.752	0.977	0.98	0.986	0.984	0.893	0.962	0.988	0.82
Z4.3	0.616	0.634	0.641	0.644	0.652	0.614	0.605	0.668	0.923
Z4.4	0.749	0.827	0.855	0.84	0.832	0.771	0.796	0.853	0.949

From the results of the cross-loading estimation, it shows that the loading value of each indicator item from the dimensions and variables to its construct is greater than the cross-loading value, namely in the Organizational Capability dimension: Export Strategy (Y2) with indicators Y2.1, Y2.7, Export Market Characteristics (Y4) with indicators Y4.2, Y4.4, Business Environment (Z1) with indicator Z1.1. This means that the construct or latent variable and its dimensions do not yet have good discriminant validity, where the indicators in the construct indicator block are not better than the indicators in other blocks.

From the results of the cross-loading analysis, the indicators of the Organizational Capability dimension: Export Strategy (Y2) with indicators Y2.1, Y2.7, Export Market Characteristics (Y4) with indicators Y4.2, Y4.4, Business Environment (Z1) with indicator Z1.1 are removed from the model. After dropping the indicators on the dimensions that did not pass the first stage of the discriminant validity test, the second stage of the discriminant validity test was carried out.

Cross-loading Value Calculation Results (Second Stage)

From the results of the cross loading estimation second stage, it shows that the loading value of each indicator item from the dimensions and variables to the construct is greater than the cross loading value, namely in the Export Knowledge-Based Factor dimension (Y3) with indicator Y3.1. This means that the construct or latent variable and its dimensions do not yet have good discriminant validity, where the indicators in the construct indicator block are not better than the indicators in other blocks. From the results of the cross loading analysis, the indicator from the Export Knowledge-Based Factor dimension (Y3) with indicator Y3.1 was removed from the model. After dropping the indicators on the dimensions that did not pass the second stage of the discriminant validity test, the third stage of the discriminant validity test was carried out.

Cross-loading Value Calculation Results (Third Stage)

From the results of the cross loading estimation third stage, it shows that the loading value of each indicator item from the dimensions and variables to its construct is greater than the cross loading value, namely in the Supply Chain dimension (Z3) with indicators Z3.2, Z3.4. This means that the construct or latent variable and its dimensions do not yet have good discriminant validity, where the indicators in the construct indicator block are not better than the indicators in other blocks. From the results of the cross loading analysis, the indicators from the Supply Chain dimension (Z3) with indicators Z3.2, Z3.4 were removed from the model. After dropping the indicators on the dimensions that did not pass the third stage of the discriminant validity test, the fourth stage of the discriminant validity test was carried out.

Cross-loading Value Calculation Results (Fourth Stage)

Indicator	Industrial Efficiency (X)	Export Performance (Y)				Competitive Strategy (Z)			
		Y1	Y2	Y3	Y4	Z1	Z2	Z3	Z4
X1	0.818	0.732	0.567	0.718	0.700	0.25	0.669	0.701	0.610
X2	0.921	0.686	0.649	0.715	0.657	0.352	0.636	0.718	0.744
X3	0.910	0.602	0.557	0.639	0.588	0.297	0.531	0.641	0.684
X4	0.916	0.597	0.632	0.63	0.575	0.348	0.569	0.631	0.692
X5	0.815	0.567	0.543	0.597	0.575	0.452	0.518	0.562	0.497
X6	0.961	0.610	0.578	0.641	0.588	0.375	0.548	0.622	0.694
X7	0.960	0.584	0.550	0.616	0.566	0.359	0.520	0.599	0.680
X8	0.908	0.590	0.574	0.621	0.565	0.383	0.534	0.606	0.678
Y1.1	0.689	0.998	0.817	0.976	0.950	0.430	0.969	0.952	0.771
Y1.2	0.686	0.996	0.864	0.988	0.940	0.443	0.972	0.960	0.781
Y1.3	0.701	0.996	0.834	0.980	0.938	0.428	0.969	0.959	0.805
Y1.5	0.686	0.998	0.837	0.982	0.949	0.438	0.972	0.956	0.789
Y2.3	0.647	0.840	1.000	0.851	0.716	0.399	0.817	0.866	0.733
Y3.2	0.689	0.981	0.806	0.989	0.973	0.457	0.979	0.954	0.788
Y3.3	0.746	0.960	0.859	0.985	0.898	0.423	0.935	0.939	0.800

Y3.4	0.707	0.982	0.861	0.996	0.946	0.457	0.974	0.969	0.815
Y4.3	0.671	0.947	0.716	0.949	1.000	0.525	0.962	0.898	0.731
Z1.2	0.388	0.436	0.399	0.451	0.525	1.000	0.463	0.364	0.395
Z2.1	0.626	0.970	0.801	0.968	0.963	0.457	0.994	0.937	0.762
Z2.3	0.629	0.966	0.823	0.965	0.949	0.464	0.994	0.930	0.743
Z3.1	0.685	0.915	0.797	0.915	0.846	0.303	0.885	0.979	0.793
Z3.3	0.702	0.965	0.898	0.973	0.912	0.408	0.954	0.981	0.810
Z4.3	0.616	0.634	0.569	0.652	0.591	0.315	0.605	0.668	0.924
Z4.4	0.749	0.827	0.786	0.846	0.764	0.415	0.796	0.849	0.948

From the results of the cross loading estimation fourth stage, it shows that the cross loading value for each indicator of each latent variable and its dimensions has been greater than the cross loading value when associated with the latent variables and dimensions reflected by other indicators. This means that each latent variable and its dimensions already have good discriminant validity where several latent variables and their dimensions have measures that are highly correlated with the construct of the latent variable and other dimensions.

Construct Reliability and Validity

Discriminant validity can also be measured by comparing the square root Average Variance Extracted (AVE) value of each construct with the correlation between other constructs in the model. The AVE value must be greater than 0.50 or have a p-value less than the 5% significance level (Hair et al., 2019: 78). The results of discriminant validity measurements in this study can be seen in the following table:

Construct Reliability and Validity

	Cronbach's Alpha	rho_A	Composite Reliability	Average Variance Extracted (AVE)
EI(X)	0.967	0.969	0.972	0.815
KE(Y)	0.992	0.993	0.994	0.945
SB(Z)	0.939	0.964	0.954	0.752
Y1	0.998	0.998	0.998	0.994
Y2	1.000	1.000	1.000	1.000
Y3	0.990	0.990	0.993	0.980
Y4	1.000	1.000	1.000	1.000
Z1	1.000	1.000	1.000	1.000
Z2	0.987	0.987	0.994	0.987
Z3	0.959	0.960	0.980	0.960
Z4	0.860	0.881	0.934	0.876

Based on the results of the Average Variance Extracted (AVE), all variables and their dimensions have discriminant validity values above 0.50. From these results, it can be concluded that all variables and their dimensions are valid and can provide confidence. After the variables and their dimensions have been declared valid, a reliability test is carried out on all variables and their dimensions used in this study.

Reliability testing is needed to measure the stability and consistency of an instrument in measuring a concept or variable. Reliability criteria can also be seen from the reliability value of a construct (Hair et al., 2019: 77). In this study, reliability can be measured by looking at the composite reliability value. To determine whether a measuring instrument is reliable or not, the reliability coefficient is used. The reliability coefficient must be greater than 0.70 (Hair et al., 2019: 77). Based on the results of the composite reliability and Cronbach alpha, all variables and their dimensions have a composite reliability value greater than 0.70. From these results, it can be concluded that all variables and their dimensions in this research model are reliable and can be relied on for use in further analysis tests.

Structural Evaluation of the Model: Second Order Model Test

Structural model or inner model testing is carried out to measure the overall relationship of latent variables in this study. Inner model measurement is carried out to determine the level of influence of the relationship between latent variables, as well as the level of influence of the overall relationship of indicators and dimensions of latent variables in the model built. In the evaluation of the structural model, all dimensions of each variable will be consolidated in the latent variable and the dimensions will be measured as indicators. This testing stage is in accordance with the second order testing procedure (variable dimensions) according to Sarstedt, et al. (2018) which explains that testing of variable dimensions must go through 2 stages of testing to accurately see the level of validity and reliability of the variable dimensions (low order) before they can be used as indicators at the level of testing the relationship between the main variables (high order). The following is a picture of a structural model that describes the relationship between dimension variables as indicators of their main variables:

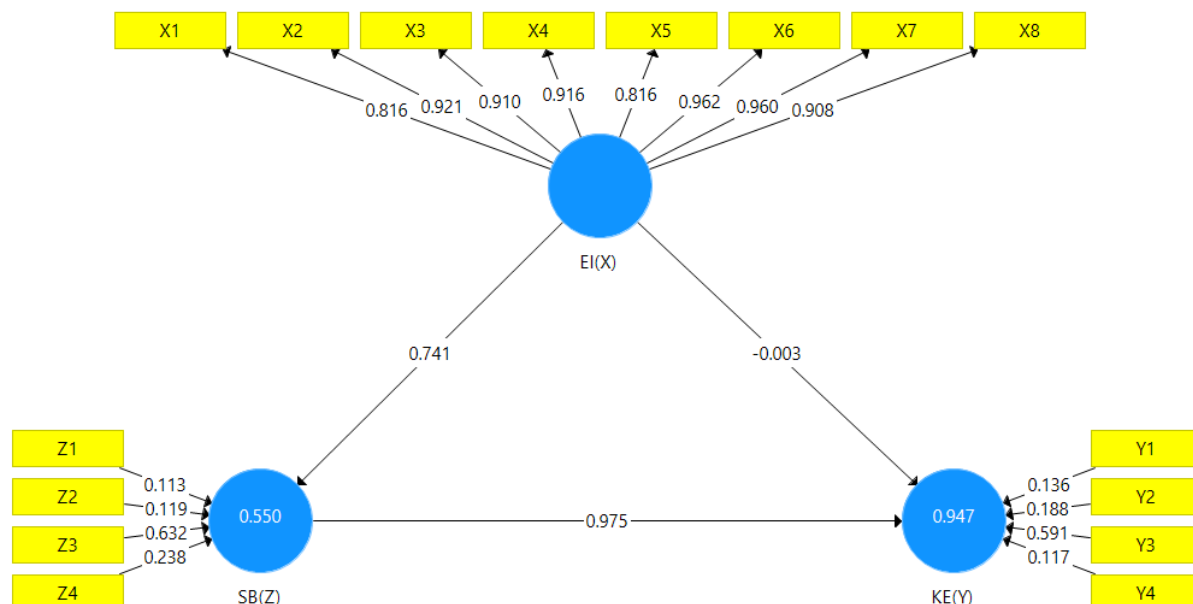


Image of second order research model and PLS model based on

However, before entering the structural model test, each main variable formed by each dimension must be retested for the level of validity testing of the formative indicators against the variables using the loading factor and indicator weight values of each formative indicator (Hair et al., 2019: 93). The expected loading factor value for testing the validity of the formative indicator is > 0.5 and the expected indicator weight value is $p\text{-value} < 0.5$ with a significance level of 5%.

Results of Measurement of Formative Indicators Validity

Variable Indicator	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values
X1 <- EI(X)	0.816	0.816	0.084	9.678	0.000
X2 <- EI(X)	0.921	0.922	0.027	34.720	0.000
X3 <- EI(X)	0.910	0.915	0.023	38.894	0.000
X4 <- EI(X)	0.916	0.917	0.039	23.666	0.000
X5 <- EI(X)	0.816	0.811	0.100	8.130	0.000
X6 <- EI(X)	0.962	0.964	0.009	106.570	0.000
X7 <- EI(X)	0.960	0.961	0.008	114.794	0.000
X8 <- EI(X)	0.908	0.917	0.025	36.532	0.000
Y1 -> KE(Y)	0.987	0.980	0.014	70.129	0.000
Y2 -> KE(Y)	0.889	0.893	0.042	21.029	0.000
Y3 -> KE(Y)	0.996	0.988	0.011	86.619	0.000
Y4 -> KE(Y)	0.941	0.929	0.064	14.787	0.000
Z1 -> SB(Z)	0.493	0.487	0.179	2.755	0.006
Z2 -> SB(Z)	0.946	0.932	0.034	28.171	0.000
Z3 -> SB(Z)	0.980	0.974	0.016	63.010	0.000
Z4 -> SB(Z)	0.890	0.888	0.045	19.789	0.000

Based on the results of the indicator loading and indicator weight tests, it can be seen that the Business Environment dimension variable (Z1) has a loading factor value of less than 0.5 so that it is not suitable for use as an indicator to reflect the Competitive Strategy variable (Z) so it must be removed from the model. This means that the Business Environment dimension variable (Z1) has a validity level that is not significant to the Competitive Strategy variable (Z). So it is necessary to carry out second stage loading indicator and weight indicator testing.

Results of Measurement of Formative Indicators Validity (Second Stage)

Indikator Variabel	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values
X1 <- EI(X)	0.817	0.815	0.084	9.767	0.000
X2 <- EI(X)	0.921	0.922	0.026	35.309	0.000
X3 <- EI(X)	0.911	0.915	0.024	38.588	0.000

X4 <- EI(X)	0.916	0.917	0.039	23.245	0.000
X5 <- EI(X)	0.815	0.810	0.100	8.157	0.000
X6 <- EI(X)	0.961	0.964	0.009	107.336	0.000
X7 <- EI(X)	0.960	0.961	0.008	115.136	0.000
X8 <- EI(X)	0.908	0.916	0.024	37.701	0.000
Y1 -> KE(Y)	0.988	0.983	0.022	44.779	0.000
Y2 -> KE(Y)	0.885	0.887	0.049	18.008	0.000
Y3 -> KE(Y)	0.997	0.991	0.010	100.774	0.000
Y4 -> KE(Y)	0.935	0.923	0.071	13.206	0.000
Z2 -> SB(Z)	0.948	0.937	0.042	22.474	0.000
Z3 -> SB(Z)	0.985	0.981	0.014	72.051	0.000
Z4 -> SB(Z)	0.895	0.894	0.045	19.969	0.000

After dropping the dimensions as indicators of the main variables that did not pass the formative indicator validity test, it can be seen in Table 9 that each dimension as an indicator of the main variable has a loading factor value of > 0.5 and the expected indicator weight value has a p-value < 0.5 with a significance level of 5%. This means that each latent variable and its dimensions already have a good level of formative indicator validity and are worthy of being used as indicators to reflect the Competitive Strategy variable (Z).

The next test is the reliability test of the formative indicators through the test of the inner multicollinearity value or VIF. Kock & Lynn (2012) proposed a full collinearity test as a comprehensive procedure for the simultaneous assessment of vertical and lateral collinearity in the direction of the relationship of the influence of formative indicators on latent variables (see, also, Kock & Gaskins, 2014). Through this procedure, which is fully automated by software, the variance inflation factors (VIFs) generated for all latent variables in the model must be free from multicollinearity. The occurrence of a VIF greater than 3.3 is proposed as an indication of pathological collinearity, and also as an indication that the model may be contaminated by common method bias (CMB). Therefore, if all VIFs generated from the full collinearity test are equal to or lower than 3.3, the model can be considered free from common method bias and is said to have no multicollinearity.

Collinearity Statistics (VIF)

Outer VIF Values		Inner VIF Values	
	EI(X)	KE(Y)	SB(Z)
EI(X)		2.161	1.000
KE(Y)			
SB(Z)		2.161	

Based on the table above, it can be concluded that there is no common method bias and no multicollinearity in the formative indicator relationship model to latent variables. This is because all values of each influence between variables do not have a value of more than 3.3 so that they

can be tested at the stage of testing the structural model of the relationship between latent variables.

Coefficient of Determination (R²)

The measurement of the inner model to test the influence between latent variables as a whole used in the study uses the R² value. R Square (R²) is often referred to as the coefficient of determination, is to measure the goodness of fit of the regression equation, namely providing the proportion or percentage of total variation in the dependent variable explained by the independent variable. The R² value lies between 0 - 1, and the model fit is said to be better if R² is getting closer to 1 (Hair et al., 2019: 118). Based on the R² value, a model can be classified as strong (≤ 0.70), medium (≤ 0.45) and weak (≤ 0.25). The following are the results of the R² value:

Quality Criteria

R Square

	R Square	R Square Adjusted
KE(Y)	0.948	0.944
SB(Z)	0.537	0.524

Based on Table, it can be seen that the R-squares result is 0.537 (53.7%) for Competitive Strategy (Z), and 0.948 (94.8%) for Export Performance (Y). This shows that the Industrial Efficiency variable (X) can affect the Competitive Strategy variable (Z) by (0.537) or 53.7%, the rest (0.463) or 46.3% is influenced by other variables outside the study. The Industrial Efficiency variable (X) can affect Export Performance (Y) by (0.948) or 94.8%, the rest (0.52) or 5.2% is influenced by other variables outside the study.

Hypothesis Testing

Final Results

Path Coefficients

Mean, STDEV, T-Values, P-Values	Confidence Intervals	Confidence Intervals Bias Corrected	Samples		
	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values
EI(X) -> KE(Y)	0.012	-0.018	0.076	0.154	0.878
EI(X) -> SB(Z)	0.733	0.750	0.051	14.502	0.000
SB(Z) -> KE(Y)	0.965	0.986	0.053	18.045	0.000

The estimated value of the path coefficient between constructs must have a significant value. The significance of the relationship can be obtained by the Bootstrapping or Jackknifing procedure. The resulting value is a t-count/t-statistic value which is then compared with the t-table. If the t-count value > t-table (1.96) at the significance level (5%) then the estimated value of the path coefficient is significant. In addition, hypothesis testing can also be done by looking at the probability value. If the p-value < 0.05 then the hypothesis is declared significant or accepted and if the p-value > 0.05 then the hypothesis is declared insignificant or rejected. This is done to strengthen the assumptions in hypothesis testing by providing alternative hypothesis testing (Hair et al., 2019: 120).

1. Industrial Efficiency Affects Export Performance

Based on the test results in the table above, the effect of Industrial Efficiency on Export Performance shows a positive coefficient value (standardized coefficient) of 0.012 and a t-Statistic of 0.154 < t-table (1.96) with a p-value of 0.878 > 0.05, then the relationship between the variables is declared insignificant or rejected. This can be interpreted that the effect of Industrial Efficiency on Export Performance is proven to have an insignificant effect.

2. Industrial Efficiency Affects Competitive Strategy

Based on the test results in the table above, the effect of Industrial Efficiency on Competitive Strategy shows a positive coefficient value (standardized coefficient) of 0.733 and a t-statistic of 14.502 > t-table (1.96) with a p-value of 0.000 < 0.05, then the relationship between the variables is declared significant or accepted. This can be interpreted that the effect of Industrial Efficiency on Competitive Strategy is proven to have a positive and significant effect.

3. Competitive Strategy Influences Export Performance

Based on the test results in the table above, the influence of Competitive Strategy on Export Performance shows a positive coefficient value (standardized coefficient) of 0.965 and a t-Statistic of 18.045 > t-table (1.96) with a p-value of 0.000 < 0.05, then the relationship between the variables is declared significant or accepted. This can be interpreted that the influence of Competitive Strategy on Export Performance has proven to have a positive and significant effect.

Mediation Testing

Specific Indirect Effects

Mean, STDEV, T-Values, P-Values	Confidence Intervals		Confidence Intervals Bias Corrected		Samples	
	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O /STDEV)	P Values	
El(X) -> SB(Z) -> KE(Y)	0.707	0.739	0.064	11.081	0.000	

Competitive Strategy Mediates the Effect of Industrial Efficiency on Export Performance

Based on the test results in the table above, the effect of the indirect relationship between Industrial Efficiency on Export Performance through Competitive Strategy as an intervening variable shows a positive coefficient value (standardized coefficient) of 0.707 and a t-Statistic of 11.081 > t-table (1.96) with a p-value of 0.000 < 0.05 the indirect variable relationship is declared significant or accepted. This can be interpreted that the indirect relationship between Industrial Efficiency on Export Performance through Competitive Strategy as an intervening variable has proven to have a positive and significant effect. So that Competitive Strategy as an intervening variable has proven to be able to mediate the indirect relationship between Industrial Efficiency on Export Performance with a positive direction of influence.